BrickPlan: Diagnosing Future Action Cost and Error Recovery in Visual Assembly Planning

Supplementary Material

A. Exact labels and admissible bounds

The archived predecessor audit evaluates every stored nonnegative state–goal label. Let R be occupied blocker closure for required removals and insertions. Without capacity or inspection obligations, the formula $|A \triangle G| + 2|R \cap A \cap G|$ agrees with all 1,802,728 audited labels. This is a finite-census equality, not a general theorem about arbitrary assembly domains or visual perception.

For BrickPlan's lower bound, initially mismatched sites each require an operation. Every occupied target site in the required removal closure must be detached and reattached. Every unfinished job needs one audit. These counts refer to disjoint required operations. Recursively including occupied path blockers can only add necessary removals. Ignoring capacity scheduling and some connectivity effects relaxes the problem, so $h(x) \leq C^*(x)$. The verifier checks this inequality at every oracle node.

For every nonterminal product node, the certificate checks $C^*(x) = \min_{a \in A(x)}[1 + C^*(T(x, a))]$, with terminal cost zero. Physical edges are compared with independent runtime transition enumeration and inverse execution; audit edges are replayed as well. Every retained product state reaches a terminal. The physical graph is reversible within its capacity-constrained component. Audits add a persistent bit without removing a physical action, and all admitted jobs can be exposed. Consequently, legal successors do not become genuine irreversible dead ends. Budget exhaustion and extra cost remain possible.

B. Construction details

The frozen candidate registry and source plan fix new kit identities, intake strata, hash-assigned source-group splits, geometry admission rules, and task sampling before qualification or solver discovery. The exact registry hash is archived. New admissions require at least eight registered and eight reachable movable sites; legacy smaller domains remain historical. Deterministic source-order endpoint pruning and bounded stud-connected extraction retain original instance identities and transforms. The predecessor's earlier pruning refinement remains disclosed in its frozen record; it is not retroactively called preregistered.

Independent original-MPD hierarchy and identity audits are an additional release gate. They exclude kits 10124, 10226, 10261, 10262 and 21318 because their part-only manifests omit direct model geometry, and kit 30312 because its original root identifies a different kit. These exclusions follow the initial geometric qualification and do not depend on policy performance. The earlier admissions and unchanged original files are archived. Every retained new-source part's index and instance ID are bound to its exact original reference chain, part/color identity and composed matrix. Failed downloads remain acquisition exclusions, without claims that the official models themselves are defective.

Each newly admitted source contributes 2 goals, 3 job settings (zero, two, and four inspections), and 2 seeded nonterminal starts per mission. The goal is either the full assembly or a seeded supported partial target. The rack holds $\lceil 0.6|V| \rceil$ pieces. Reachability, not cost, gap, policy performance, or expansion count, admits tasks. Easy episodes remain included. Geometry contrasts are a separate mechanism-conditioned diagnostic, not part of this source-sampling factorial.

Primary matched contexts are induced by deleting one or two original fixed instances from each side. Geometry-only enumeration groups deletions by exact part-number/color inventory, keeps distinct support masks, and selects at most four context pairs per base in seeded hash order. It never moves pieces, adds spacers, substitutes meshes, or falls back to unequal counts. Both fixed structures remain connected and the full assemblies supported; endpoint and straight-path qualification are recomputed. Registered instances, poses, start, goal, capacity, jobs, and proposed plan match.

After this selection, a bounded differing-edge search seeks exact optimal completions with a shared legal prefix of at least two operations. At most two distinct starting questions are retained per intervention. Of 116 qualified primary interventions, 98 yield no retained pair. This is search-specific attrition, not proof that other contrasts cannot exist. The retained 36 pairs and 10 unmatched historical regressions have separate denominators. Original/grayscale controls preserve all geometry and labels.

The 216 interior-order pairs are balanced over conflict bins 2–5, 6–10, and 11+. Legal and reordered plans share exact prefixes up to the first conflict, endpoints, and action multisets. Legal witnesses have no repeated states. The stored audit exposes each conflict position, so readers can run other prefix controls.

C. Shallow choice and control diagnostics

The depth-one minimizing set contains an optimal action at 2,595 of 2,595 selected checkpoints. These privileged struc-

Table 1. A prefix-only software control returns the first conflict if it appears within k executed operations, otherwise "all legal." The denominator is all 432 continuation questions.

Prefix length	Correct / assigned	Accuracy
0	216/432	50.0%
1	216/432	50.0%
2	234/432	54.2%
3	257/432	59.5%
5	288/432	66.7%
8	340/432	78.7%
11	372/432	86.1%
16	410/432	94.9%
24	426/432	98.6%
32	432/432	100.0%

Table 2. Measured finite-horizon choice sets for all 2,595 next-action questions. Any/all refer to optimistic/pessimistic ties; uniform tie correctness is an exact average, not a sampled policy result.

Depth	Choices	All optimal	Any optimal	Uniform tie
1	2595	2445	2595	97.4%
2	2595	2484	2595	98.1%
4	2595	2505	2595	98.3%
8	2595	2544	2595	99.0%

tural choices do not establish a required internal reasoning depth for image-input policies. Deeper horizons can change tie composition nonmonotonically. The full table is retained without redefining difficulty to match its results.

All 39,788 raw policy task-run records are archived in `data/software-control-records.json.gz`. They include one oracle seed and three seeds for nine other policies, with actions, outcomes, expansions, timings, and regrets. The release checker recomputes every overall, tier, source, and seed aggregate. The full search cap of 500,000 is a resource ceiling, not actual expansion usage. Oracle loading and cached local timing are not comparable to calibrated online search.

D. Recovery, diagnosis, and censoring

Traverse only edges with $C^*(T(x,a)) = C^*(x) - 1$ from pre-error state b. The reached states are exactly the union of optimal completions: each traversal is an optimal prefix, and every optimal prefix satisfies that equality. Remove the terminal state to obtain the repair target $\mathcal{C}_b^-$. The generator's delayed checkpoints remain outside this set after an error and lag 1, 3, or 5. Independent forward BFS verifies their shortest return distances, which range from 1 to 13. Lag-specific denominators for the 3,829 questions appear below.

Each positive-excess legal action creates its own spontaneous event; later actions update every active event. A second error never replaces the first. Immediate recovery and lag-specific delayed cohorts are reported separately. Unfinished events are censored; terminal events that never restore a nonterminal corridor state are observed failures. Reaching a goal after a costly detour earns completion credit, but not automatic corridor-repair credit.

The inverse-last-action control uses only the public executed history and candidate list. It earns zero when no inverse is offered. Its combined 2,260/3,829 accuracy is 59.0%. There are 1,487 cases with a legal but suboptimal inverse and 82 with no offered inverse. Lag is thus an experimental descriptor, not proof of necessary nonlocal reasoning.

There are 50 all-correct-choice calibration questions across the diagnostic bank. Their correct tie sets are preserved, and the evaluator reports calibration separately from the 6,970 discriminative rows. The 7,020 rows map to 6,830 canonical problems. Assigned repeated variants are averaged before problems and sources; missing responses receive zero before this averaging. The evaluator exposes both row and clustered scores, plus each inverse-control cohort. None of these private labels is included in the policy observation.

An accepted diagnosis concerns legal positive-excess events since the last diagnosis or episode start. The first later diagnosis is attached to each previously undiagnosed event, with its sign and lag. True positive diagnosis checkpoints open replanning-suffix records. Always-negative or selectively requested diagnoses can yield misleading accuracy, so report event detection and query coverage alongside it.

Plan-survival reports retain the first cost loss even after recovery, and include explicit eligible/observed/unknown counts per operation index. They are descriptive task-budget curves, not Kaplan–Meier population estimates. Conditions should use matched assignments and registered source strata. Missing episodes must not disappear from denominators.

D.1. Regret conservation

For a successful trace, accepted attaches A_c and detaches D_c satisfy $A_c - D_c = \Delta$, the target-minus-start occupancy count. There are exactly u accepted previously unfinished audits. Duplicate audits are rejected. If N_{invalid} is the number of charged invalid attempts, then

$$T = 2D_c + \Delta + u + N_{\text{invalid}}, \tag{1}$$

$$C^*(x_0) = 2D^* + \Delta + u, \tag{2}$$

$$T - C^*(x_0) = 2(D_c - D^*) + N_{\text{invalid}}. \tag{3}$$

This audit-inclusive identity reconciles detachment regret and total excess. Rejected actions consume budget; stale

Table 3. Structural policy completion by operation-length bin. Each nonoracle policy has three tie-order seeds per episode.

Control	D1	D2	D3	D4
Closure greedy	942/942	2096/2103	988/1023	137/195
Capacity dispatch	942/942	2097/2103	970/1023	123/195
Closure A* (100)	942/942	2088/2103	946/1023	99/195
Closure A* (1,000)	942/942	2103/2103	1005/1023	149/195
Closure A* (full)	942/942	2103/2103	1023/1023	195/195
Hamming A* (full)	942/942	2103/2103	1023/1023	195/195

Table 4. Delayed-repair lag and public inverse-last control.

Lag	Questions	Inverse correct	Accuracy
1	1309	919	70.2%
3	1268	715	56.4%
5	1252	626	50.0%

Table 5. Recovery cohorts are not pooled into a claim of nonlocal repair.

Inverse cohort	Questions	Sources	Repair cost
Optimal inverse	2260	97	2–13
Suboptimal inverse	1487	96	1–7
No offered inverse	82	44	1–11

HTTP revisions and malformed request envelopes are rejected before mutation and are not action events.

E. Visual stimuli and actual model calls

All executable geometry uses original source poses and LDraw triangle meshes. The 449 distinct part meshes contain 523,571 triangles. `ASSET-LICENSES.json` preserves source models, authors, URLs, and part-level notices; attribution and licenses remain separate from code. Qualification proxies do not replace the displayed shapes. Figure provenance records exact states, seeds, bank hash, and image hashes.

The public renderer supersamples by two before Lanczos downsampling to native HD. Display canvases account for device-pixel ratio, and native zoom avoids CSS-only enlargement of low-resolution pixels. Catalog tiles are 480×360 and catalogs are 1920 pixels wide. Models receive image content, not text describing images. Preprocessing is separately declared: the local SmolVLM case resizes to longest edge 512, with splitting disabled. HD display does not imply native-HD model processing.

The preserved version-2.1 local case contains four actual inference calls on historical public E069, seed 0. The original task and bank hash are archived; this trace is not relabeled as version-2.3 evaluation. All calls return descriptive prose instead of an executable JSON action. The executor rejects each, and the controller stops at its four-call limit. Every prompt, raw response, image hash, latency, measured token count, action, and feedback is archived. No oracle action or fabricated explanation replaces model output. This is a failed integration example. It supplies neither an estimate of SmolVLM accuracy nor evidence about larger VLMs.

F. Protocol and reproducibility

Study runs bind participant, condition, seed, assignments, and bank hash. The evaluator replays mutations, validates views, and normalizes candidate permutations. Retries are idempotent across restart; changed content or stale revisions are rejected. Failed atomic persistence rolls back mutations.

Practice exposes rich catalogs, solution review after submission, and free 3D inspection. Study mode blocks these and private downloads. Portable evaluator data must remain on the evaluator host during blind studies; a local working-directory convention is not a security sandbox.

The practice model manager supports local MLX and OpenAI-compatible vision calls, durable outputs, and bounded run/step/stop. Credentials never enter browser exports. Stop prevents subsequent actions, not in-flight provider billing. Interrupted calls are never silently retried.

The population study runner is distinct. Its reported-usage mode reserves maximum tokens and cost before calls; adapters account for images and attest usage. Missing accounting, timeout, malformed responses, or limit violations stop continuation. Unavailable-accounting mode marks tokens and currency unavailable. This relies on adapter/provider reports, not a provider-enforced spending quota.

The main paper uses the unmodified official CVPR author-kit, pinned to commit `2917585`, labeled 2026. A future year's kit is not claimed. The PDF uses public preprint mode without margin/font compression. Review mode requires the actual year and paper ID.

Table 6. Metric denominators and treatment of missing outcomes.

Metric	Eligible denominator	Missing / termination rule
Completion / optimal completion	All assigned episodes	Unstarted and unfinished are failures
Successful mean regret	Successfully finished episodes	Null for failures; report success coverage
Avoidable-cost choice	Legal decisions with zero- and positive-excess alternatives	Invalid attempts excluded here, charged in total
Budget-loss choice	Viable checkpoints with viable and nonviable legal choices	Other budget losses remain in raw records
True-unreachability choice	No eligible choices in retained components	Undefined, not empirical evidence of zero risk
Recovery bounds	Every eligible legal positive-excess event	Lower bound restored/all; upper adds censored
Diagnosis accuracy	Accepted diagnosis queries	Unqueried intervals are not inferred correct
Primary geometry joint correctness	Pairs with at least one assigned member	Both exact labels required; absent partner scores zero
Discriminative probe accuracy	Assigned noncalibration questions of that family	Missing or invalid response gets no credit
Source macro accuracy	Variants within problem, then problems within source, then sources	Missing variants enter as zero before averaging
Calibration accuracy	Assigned all-correct-choice questions	Report separately; not evidence of discrimination

G. Base source assemblies

The table lists 98 base assemblies; roadster and kart share kit 31027. An additional 314 qualified context profiles are indexed in the machine-readable source census. They are not independent sources. Profiles without episodes are zero-yield candidates, not benchmark tasks. There are 270 profiles without episodes. Registered and movable counts describe different quantities; "Parts" counts the admitted subassembly, not the full original kit. The active bank covers 97 source groups.

Admitted subassembly	Kit	Split	Parts	Sites	Movable	Episodes
Roadster body	31027	Dev.	37	17	17	14
Kart deck	31027	Dev.	25	18	18	14
Airframe	31028	Dev.	24	21	21	14
Truck cab and roof	10156	Dev.	76	18	18	10
House roof assembly	31009	Dev.	221	18	17	14
Sports car upper body	5867	Dev.	149	22	10	3
Source 10001 subassembly	10001	Dev.	17	2	2	8
Source 10159 subassembly	10159	Dev.	177	16	16	12
Source 10220 subassembly	10220	Dev.	153	16	16	12
Source 10014 subassembly	10014	Held out	71	16	15	12
Source 10128 subassembly	10128	Held out	98	16	14	12
Source omr-31031 subassembly	31031	Held out	42	12	12	12
Source 10036 subassembly	10036	Held out	79	16	16	48
Source 10213 subassembly	10213	Held out	179	2	2	28
Source omr-10242 subassembly	10242	Held out	180	16	16	48
Source omr-10265 subassembly	10265	Held out	179	4	4	48
Source omr-10269 subassembly	10269	Held out	105	16	11	48
Breezeway Cafe	10037	Held out	65	16	12	12
Town Plan	10184	Held out	96	16	16	12
Gingerbread House	10267	Held out	96	16	16	12
Holiday Home	1472	Held out	89	16	14	12
Weetabix Town House	1484	Held out	96	16	16	12
Town Bank	1490	Held out	92	16	16	12

Admitted subassembly	Kit	Split	Parts	Sites	Movable	Episodes
House with Roof-Windows (Velux)	1854	Dev.	92	16	16	12
The Big Bang Theory	21302	Held out	96	16	15	12
Central Perk	21319	Held out	96	16	16	12
Medieval Blacksmith	21325	Held out	95	16	16	12
Mountain Hut	31025	Held out	95	16	16	12
Changing Seasons	31038	Dev.	94	16	16	12
Lakeside Lodge	31048	Dev.	96	16	16	12
Beachside Vacation	31063	Dev.	95	16	16	12
Burger Stand	6683	Held out	50	16	16	12
Post-Station	6689	Held out	39	16	16	12
Cycle Fix-It Shop	6699	Held out	44	16	16	12
Solomon R. Guggenheim Museum	21004	Held out	95	16	16	12
Fallingwater	21005	Held out	96	16	16	12
The White House	21006	Dev.	96	16	16	12
Rockefeller Center	21007	Dev.	46	16	16	12
Farnsworth House	21009	Held out	94	16	16	12
Robie House	21010	Held out	87	16	16	12
Brandenburg Gate	21011	Dev.	91	16	16	12
Sydney Opera House	21012	Dev.	48	16	16	12
Villa Savoye	21014	Held out	96	16	16	12
Sungnyemun	21016	Held out	95	16	16	12
Imperial Hotel	21017	Held out	96	16	16	12
United Nations Headquarters	21018	Dev.	60	16	16	12
The Trevi Fountain	21020	Held out	96	16	16	12
Lincoln Memorial	21022	Held out	96	16	16	12
Louvre	21024	Held out	96	16	16	12
Buckingham Palace	21029	Held out	76	16	16	12
United States Capitol Building	21030	Held out	92	16	16	12
LEGO House	21037	Held out	96	16	16	12
Trafalgar Square	21045	Dev.	83	16	16	12
Alligator	20015	Dev.	13	12	11	12
Clown Fish	30025	Held out	29	16	16	12
Panda	30026	Dev.	15	14	14	12
Reindeer	30027	Dev.	25	16	15	12
Pudsey Bear	30029	Held out	64	16	16	12
Cute Pug	30542	Held out	26	16	16	12
Pelican	30571	Dev.	25	16	16	12
Space Shuttle	1682	Held out	30	12	11	12
Jet	1775	Held out	44	15	15	12
Light Aircraft and Ground Support	1808	Held out	27	16	16	12
International Space Station	21321	Held out	91	16	16	12
Helicopter	30181	Held out	27	8	8	12
Swamp Police Helicopter	30311	Held out	27	16	15	12
Blue Power Jet	31039	Dev.	93	16	16	12
Twin Spin Helicopter	31049	Held out	87	15	15	12
Space Shuttle Explorer	31066	Held out	93	16	16	12
Helicopter Adventure	31092	Dev.	60	16	16	12
Passenger Plane	3181	Dev.	11	10	10	12
Propellor Power	6745	Held out	94	16	16	12
WALL•E [Original Version]	21303	Held out	33	14	11	12
Mobile Car Crane	1489	Held out	93	8	8	12
Super Tow Truck	1572	Dev.	63	16	16	12
Loader Tractor	1633	Dev.	13	12	9	12
Pirate Roller Coaster	31084	Held out	96	12	12	12
Repair Truck	3179	Dev.	60	16	16	12
Pneumatic Crane	6678	Dev.	16	15	12	12

Admitted subassembly	Kit	Split	Parts	Sites	Movable	Episodes
Cement Mixer	6682	Held out	20	16	16	12
Snorkel Pumper	6690	Held out	25	16	16	12
Ferrari F40	10248	Held out	92	12	11	12
London Bus	10258	Held out	9	8	8	12
Fiat 500	10271	Dev.	33	8	8	12
Adventure Vehicles	31037	Held out	96	16	16	12
Desert Racers	31040	Dev.	49	16	16	12
Turbo Track Racer	31070	Dev.	96	16	16	12
Extreme Engines	31072	Held out	56	12	12	12
LEGO Truck	3221	Held out	91	16	16	12
Highway Transport	6753	Held out	96	16	16	12
High Speed Train Car	10158	Dev.	50	8	8	12
Emerald Night	10194	Held out	90	12	12	12
Horizon Express	10233	Held out	96	16	16	12
Maersk Line Triple-E	10241	Held out	96	16	16	12
Ocean Explorer	31045	Held out	96	16	16	12
Ship	4018	Held out	9	8	8	12
RV with Speedboat	6698	Held out	36	16	13	12
Motor Boat	1632	Dev.	11	8	8	12

Partition	Sources	Episodes	Greedy	Full A*
Development	34	445	95.9%	100.0%
Held-out source	63	976	98.5%	100.0%

Version 2.3 admits 81 of 100 new frozen candidates. All 19 rejected candidates remain documented. All earlier episode and diagnostic definitions are preserved, without promoting historical unmatched pairs to primary matched evidence.

H. Source strata and admission attrition

Frozen intake stratum	Candidates	Admitted	Rejected	Held out	Pairs	Pair sources
aircraft-and-space	16	12	4	9	0	0
animals	11	7	4	3	0	0
architecture	20	19	1	13	2	1
buildings-and-scenes	20	17	3	13	18	5
mechanisms-and-working-vehicles	12	9	3	5	2	1
rail-and-water	10	8	2	6	4	1
road-vehicles	11	9	2	6	0	0

Strata are frozen intake categories, not post hoc difficulty labels. "Held out" counts admitted new sources assigned to the held-out split; "Pairs" and "Pair sources" describe retained primary geometry only. The following candidate census includes every frozen admission outcome. Exact rejection reasons, pruned instances, qualification attempts, original source URLs and authors, and dependency hashes remain in `source-plan.json`, `data/source-admission.json`, and `data/source-question-audit.json`.

Model	Kit	Split	Frozen stratum	Admission
v23-10037	10037	Held out	buildings-and-scenes	Admitted
v23-10184	10184	Held out	buildings-and-scenes	Admitted
v23-10267	10267	Held out	buildings-and-scenes	Admitted
v23-1472	1472	Held out	buildings-and-scenes	Admitted
v23-1484	1484	Held out	buildings-and-scenes	Admitted
v23-1490	1490	Held out	buildings-and-scenes	Admitted
v23-1854	1854	Dev.	buildings-and-scenes	Admitted
v23-21302	21302	Held out	buildings-and-scenes	Admitted

Model	Kit	Split	Frozen stratum	Admission
v23-21318	21318	Held out	buildings-and-scenes	Rejected
v23-21319	21319	Held out	buildings-and-scenes	Admitted
v23-21325	21325	Held out	buildings-and-scenes	Admitted
v23-31025	31025	Held out	buildings-and-scenes	Admitted
v23-31038	31038	Dev.	buildings-and-scenes	Admitted
v23-31048	31048	Dev.	buildings-and-scenes	Admitted
v23-31063	31063	Dev.	buildings-and-scenes	Admitted
v23-31153	31153	Held out	buildings-and-scenes	Rejected
v23-6683	6683	Held out	buildings-and-scenes	Admitted
v23-6689	6689	Held out	buildings-and-scenes	Admitted
v23-6699	6699	Held out	buildings-and-scenes	Admitted
v23-30023	30023	Held out	buildings-and-scenes	Rejected
v23-21004	21004	Held out	architecture	Admitted
v23-21005	21005	Held out	architecture	Admitted
v23-21006	21006	Dev.	architecture	Admitted
v23-21007	21007	Dev.	architecture	Admitted
v23-21009	21009	Held out	architecture	Admitted
v23-21010	21010	Held out	architecture	Admitted
v23-21011	21011	Dev.	architecture	Admitted
v23-21012	21012	Dev.	architecture	Admitted
v23-21014	21014	Held out	architecture	Admitted
v23-21016	21016	Held out	architecture	Admitted
v23-21017	21017	Held out	architecture	Admitted
v23-21018	21018	Dev.	architecture	Admitted
v23-21020	21020	Held out	architecture	Admitted
v23-21021	21021	Dev.	architecture	Rejected
v23-21022	21022	Held out	architecture	Admitted
v23-21024	21024	Held out	architecture	Admitted
v23-21029	21029	Held out	architecture	Admitted
v23-21030	21030	Held out	architecture	Admitted
v23-21037	21037	Held out	architecture	Admitted
v23-21045	21045	Dev.	architecture	Admitted
v23-20015	20015	Dev.	animals	Admitted
v23-30022	30022	Held out	animals	Rejected
v23-30025	30025	Held out	animals	Admitted
v23-30026	30026	Dev.	animals	Admitted
v23-30027	30027	Dev.	animals	Admitted
v23-30029	30029	Held out	animals	Admitted
v23-30285	30285	Held out	animals	Rejected
v23-30472	30472	Held out	animals	Rejected
v23-30542	30542	Held out	animals	Admitted
v23-30571	30571	Dev.	animals	Admitted
v23-3223	3223	Held out	animals	Rejected
v23-10124	10124	Dev.	aircraft-and-space	Rejected
v23-10226	10226	Dev.	aircraft-and-space	Rejected
v23-1682	1682	Held out	aircraft-and-space	Admitted
v23-1775	1775	Held out	aircraft-and-space	Admitted
v23-1808	1808	Held out	aircraft-and-space	Admitted
v23-21321	21321	Held out	aircraft-and-space	Admitted
v23-30181	30181	Held out	aircraft-and-space	Admitted
v23-30311	30311	Held out	aircraft-and-space	Admitted
v23-31039	31039	Dev.	aircraft-and-space	Admitted
v23-31049	31049	Held out	aircraft-and-space	Admitted
v23-31066	31066	Held out	aircraft-and-space	Admitted
v23-31092	31092	Dev.	aircraft-and-space	Admitted
v23-3178	3178	Dev.	aircraft-and-space	Rejected

Model	Kit	Split	Frozen stratum	Admission
v23-3181	3181	Dev.	aircraft-and-space	Admitted
v23-6745	6745	Held out	aircraft-and-space	Admitted
v23-6912	6912	Dev.	aircraft-and-space	Rejected
v23-10261	10261	Held out	mechanisms-and-working-vehicles	Rejected
v23-21303	21303	Held out	mechanisms-and-working-vehicles	Admitted
v23-21323	21323	Held out	mechanisms-and-working-vehicles	Rejected
v23-1489	1489	Held out	mechanisms-and-working-vehicles	Admitted
v23-1572	1572	Dev.	mechanisms-and-working-vehicles	Admitted
v23-1633	1633	Dev.	mechanisms-and-working-vehicles	Admitted
v23-30312	30312	Held out	mechanisms-and-working-vehicles	Rejected
v23-31084	31084	Held out	mechanisms-and-working-vehicles	Admitted
v23-3179	3179	Dev.	mechanisms-and-working-vehicles	Admitted
v23-6678	6678	Dev.	mechanisms-and-working-vehicles	Admitted
v23-6682	6682	Held out	mechanisms-and-working-vehicles	Admitted
v23-6690	6690	Held out	mechanisms-and-working-vehicles	Admitted
v23-10248	10248	Held out	road-vehicles	Admitted
v23-10252	10252	Held out	road-vehicles	Rejected
v23-10258	10258	Held out	road-vehicles	Admitted
v23-10262	10262	Held out	road-vehicles	Rejected
v23-10271	10271	Dev.	road-vehicles	Admitted
v23-31037	31037	Held out	road-vehicles	Admitted
v23-31040	31040	Dev.	road-vehicles	Admitted
v23-31070	31070	Dev.	road-vehicles	Admitted
v23-31072	31072	Held out	road-vehicles	Admitted
v23-3221	3221	Held out	road-vehicles	Admitted
v23-6753	6753	Held out	road-vehicles	Admitted
v23-10158	10158	Dev.	rail-and-water	Admitted
v23-10194	10194	Held out	rail-and-water	Admitted
v23-10233	10233	Held out	rail-and-water	Admitted
v23-10241	10241	Held out	rail-and-water	Admitted
v23-1792	1792	Held out	rail-and-water	Rejected
v23-31045	31045	Held out	rail-and-water	Admitted
v23-4018	4018	Held out	rail-and-water	Admitted
v23-6679	6679	Dev.	rail-and-water	Rejected
v23-6698	6698	Held out	rail-and-water	Admitted
v23-1632	1632	Dev.	rail-and-water	Admitted

I. Matched geometry cohorts and shortcut controls

Cohort	Split	Pairs	Interventions	Starts	Sources
Primary matched	Dev.	12	6	12	3
Primary matched	Held out	24	12	24	9
Historical unmatched	Dev.	4	1	4	1
Historical unmatched	Held out	6	3	6	2

Pairs are repeated questions nested within distinct starts, interventions, and sources. These units must not be counted as independent source samples. The primary score averages repeated rows within starts, starts within interventions, interventions within sources, and sources equally. Both exact one-based first-conflict labels (or `null`) must be correct. A one-sided assignment remains eligible and earns zero joint credit. Missing and invalid answers cannot disappear from the denominator. Wholly unassigned pairs are census-only. Historical unmatched regressions are always separate, including in held-out reports.

Representation	Split	Pairs	Equal	Member bound	Joint bound	Status
Counts	Dev.	12	12	50.0%	0.0%	Exact
Counts	Held out	24	24	50.0%	0.0%	Exact
Type/color presence	Dev.	12	12	50.0%	0.0%	Exact
Type/color presence	Held out	24	24	50.0%	0.0%	Exact
Type/color counts	Dev.	12	12	50.0%	0.0%	Exact
Type/color counts	Held out	24	24	50.0%	0.0%	Exact
Sequence	Dev.	12	12	50.0%	0.0%	Exact
Sequence	Held out	24	24	50.0%	0.0%	Exact
Public text (normalized IDs)	Dev.	12	12	50.0%	0.0%	Exact
Public text (normalized IDs)	Held out	24	24	50.0%	0.0%	Exact
Public text + inventories	Dev.	12	12	50.0%	0.0%	Exact
Public text + inventories	Held out	24	24	50.0%	0.0%	Exact

These are label-aware finite-bank deterministic information ceilings, not trained predictors or visual-model results. The member bound uses source/intervention/start weights; the joint bound optimizes a single classifier over representation-equivalence classes. When enumeration cannot establish tightness, the report retains an upper bound and a separate achievable witness rather than claiming an exact maximum. "Equal inputs" counts pairs identical under the named projection. Type/color presence is an unordered global set, not object-sensitive spatial presence. Inventory access assumes perfect extraction. Address-normalized text removes only the declared routing fields; the full-address certificate remains in `data/matched-controls.json`.

Software rule	Split	Pairs	Sources	Joint micro	Source macro	95% interval
Named-occupancy inventory	Dev.	12	3	0.0%	0.0%	[0.0, 0.0]
Named-occupancy inventory	Held out	24	9	0.0%	0.0%	[0.0, 0.0]
Public-text inventory	Dev.	12	3	0.0%	0.0%	[0.0, 0.0]
Public-text inventory	Held out	24	9	0.0%	0.0%	[0.0, 0.0]
Development source majority	Held out	24	9	0.0%	0.0%	[0.0, 0.0]

Label-free rules optimistically execute occupancy, rack, registration, and work-order constraints without querying meshes, paths, support, or exact costs. A `null` prediction means no nongeometric violation was found, not a proof of geometric legality. Source majority is fitted only on primary development labels; unseen held-out sources use its frozen development fallback. It is not fitted on held-out outcomes. The complete predictions, missing counts, per-source results, and leave-one-source-out values are released. Bootstrap intervals resample whole retained sources, not rows, interventions, or identity seeds. Fewer than two assigned sources produces an undefined interval. These are conditional finite-bank sensitivity summaries, not calibrated population guarantees.

Primary stratum	Pairs	Sources	Prefix range	Plan range
Conflict 2–5	28	10	2–4	7–26
Conflict 6–10	8	6	5–6	13–22
Conflict 11+	0	0	—	—

Conflict-position strata characterize retained questions; they are not source admission criteria or necessary reasoning-depth levels. All retained primary illegal members fail through unsupported assembly. Their motion-blocker relations match those of the paired legal contexts, so this cohort tests support connectivity rather than changed path clearance.

J. Context selection and zero-yield interventions

"Subsets" counts enumerated original fixed-instance deletions; "Matched options" counts inventory-matched options after connectivity/support filtering. "Qualified" counts selected and requalified interventions, "Yielding" those with a retained answer change, and "Pairs" retained starting questions. Dashes indicate a base excluded before enumeration, not zero evaluated candidates. Full reason counts and selected original indices are in the geometry census. No failed source or intervention is silently relabeled a successful task.

Base profile	Subsets	Matched options	Qualified	Yielding	Pairs
roadster-body	210	1	1	0	0
kart-deck	28	0	0	0	0
airframe	6	2	2	0	0
truck-cab	1711	69	4	0	0
house-roof	20706	115	4	1	2
sports-body	8128	38	1	0	0
source-10001	—	—	0	0	0
source-10159	13041	277	4	0	0
source-10220	9453	40	4	0	0
source-10014	1540	1	1	1	2
source-10128	3403	26	1	1	2
source-omr-31031	465	42	4	0	0
source-10036	2016	25	3	2	4
source-10213	—	—	0	0	0
source-omr-10242	13530	6	4	0	0
source-omr-10265	—	—	0	0	0
source-omr-10269	4005	0	0	0	0
source-v23-10037	1225	1	1	1	2
source-v23-10184	3240	0	0	0	0
source-v23-10267	3240	1	1	0	0
source-v23-1472	2701	62	4	1	2
source-v23-1484	3240	30	1	0	0
source-v23-1490	2926	160	4	0	0
source-v23-1854	2926	179	4	0	0
source-v23-21302	3240	0	0	0	0
source-v23-21319	3240	0	0	0	0
source-v23-21325	3160	0	0	0	0
source-v23-31025	3160	24	3	1	2
source-v23-31038	3081	0	0	0	0
source-v23-31048	3240	59	4	4	8
source-v23-31063	3160	1	1	0	0
source-v23-6683	595	37	4	2	4
source-v23-6689	276	63	4	0	0
source-v23-6699	406	10	2	0	0
source-v23-21004	3160	1	1	0	0
source-v23-21005	3240	0	0	0	0
source-v23-21006	3240	0	0	0	0
source-v23-21007	465	0	0	0	0
source-v23-21009	3081	36	4	0	0
source-v23-21010	2556	0	0	0	0
source-v23-21011	2850	0	0	0	0
source-v23-21012	528	0	0	0	0
source-v23-21014	3240	0	0	0	0
source-v23-21016	3160	88	4	0	0
source-v23-21017	3240	0	0	0	0
source-v23-21018	990	11	1	1	2
source-v23-21020	3240	0	0	0	0
source-v23-21022	3240	0	0	0	0
source-v23-21024	3240	0	0	0	0
source-v23-21029	1830	0	0	0	0
source-v23-21030	2926	0	0	0	0
source-v23-21037	3240	0	0	0	0
source-v23-21045	2278	0	0	0	0
source-v23-20015	1	0	0	0	0
source-v23-30025	91	5	4	0	0
source-v23-30026	1	0	0	0	0

Base profile	Subsets	Matched options	Qualified	Yielding	Pairs
source-v23-30027	45	0	0	0	0
source-v23-30029	1176	16	3	0	0
source-v23-30542	55	3	1	0	0
source-v23-30571	45	0	0	0	0
source-v23-1682	171	0	0	0	0
source-v23-1775	435	0	0	0	0
source-v23-1808	66	4	1	0	0
source-v23-21321	2850	1	1	0	0
source-v23-30181	190	0	0	0	0
source-v23-30311	66	0	0	0	0
source-v23-31039	3003	65	4	0	0
source-v23-31049	2628	0	0	0	0
source-v23-31066	3003	24	1	0	0
source-v23-31092	990	0	0	0	0
source-v23-3181	1	0	0	0	0
source-v23-6745	3081	0	0	0	0
source-v23-21303	190	0	0	0	0
source-v23-1489	3655	0	0	0	0
source-v23-1572	1128	0	0	0	0
source-v23-1633	1	0	0	0	0
source-v23-31084	3570	78	4	1	2
source-v23-3179	990	16	1	0	0
source-v23-6678	1	0	0	0	0
source-v23-6682	10	0	0	0	0
source-v23-6690	45	0	0	0	0
source-v23-10248	3240	0	0	0	0
source-v23-10258	1	0	0	0	0
source-v23-10271	325	0	0	0	0
source-v23-31037	3240	0	0	0	0
source-v23-31040	561	24	3	0	0
source-v23-31070	3240	0	0	0	0
source-v23-31072	990	125	4	0	0
source-v23-3221	2850	0	0	0	0
source-v23-6753	3240	68	4	0	0
source-v23-10158	903	0	0	0	0
source-v23-10194	3081	0	0	0	0
source-v23-10233	3240	66	3	0	0
source-v23-10241	3240	9	4	2	4
source-v23-31045	3240	1	1	0	0
source-v23-4018	1	0	0	0	0
source-v23-6698	210	0	0	0	0
source-v23-1632	6	1	1	0	0

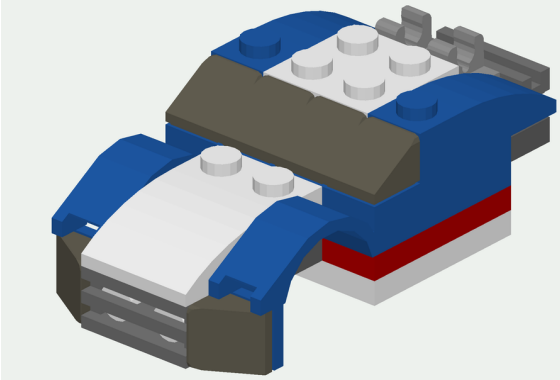
K. Static and active view certificates

Static diagnostics supply four fixed views at revision zero: isometric, left, rear, and right. All five panels are included for each view, and later acquisition is disabled. The structured diagnostic condition uses the same initial views. Active image diagnostics begin with isometric only; each inspect increments the view budget. These are separate conditions rather than interchangeable names for one observation.

The native-resolution certificate compares every geometry pair in all twenty initial panels. Pair-specific revealing views and isometric-only ambiguities are listed in `data/observation-certificate.json`, with primary and historical membership inherited from the bank. The certificate is bound to the bank hash and identity seed. It demonstrates input distinguishability, not human agreement or perceptual answerability. Other image-processing pipelines and identity seeds require their own checks. Study results must report preprocessing and acquisition counts.

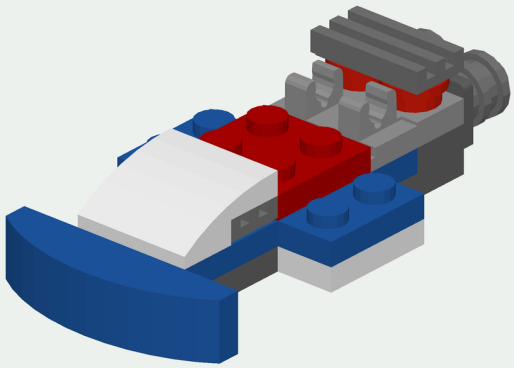
Original-source gallery: assemblies 1–4

roadster-body | development



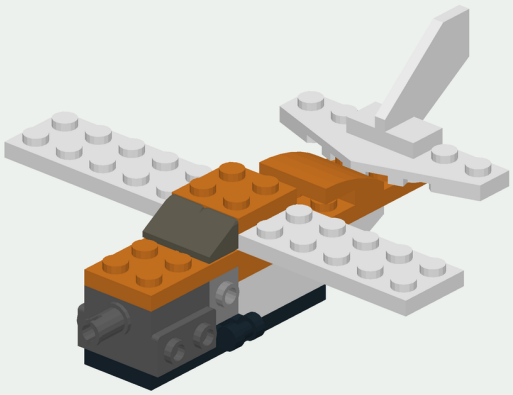
Kit 31027; 37 parts; 17 registered
17 reachable movable sites

kart-deck | development



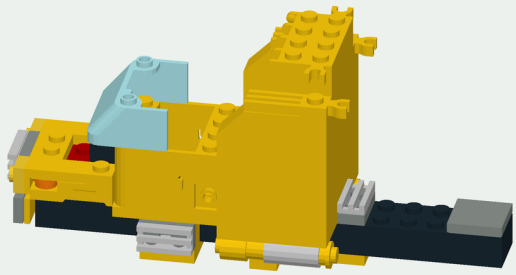
Kit 31027; 25 parts; 18 registered
18 reachable movable sites

airframe | development



Kit 31028; 24 parts; 21 registered
21 reachable movable sites

truck-cab | development



Kit 10156; 76 parts; 18 registered
18 reachable movable sites

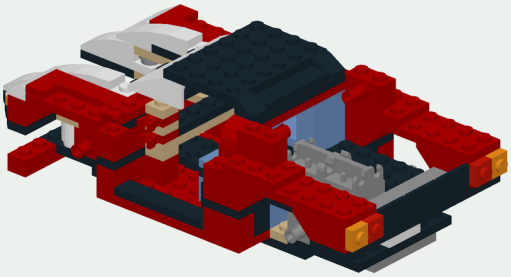
Original-source gallery: assemblies 5–8

house-roof | development



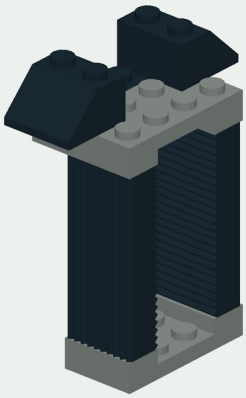
Kit 31009; 221 parts; 18 registered
17 reachable movable sites

sports-body | development



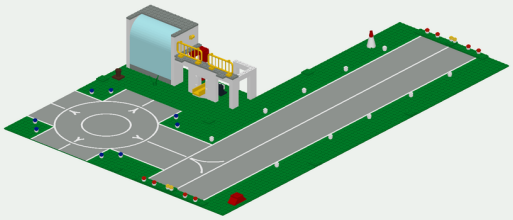
Kit 5867; 149 parts; 22 registered
10 reachable movable sites

source-10001 | development



Kit 10001; 17 parts; 2 registered
2 reachable movable sites

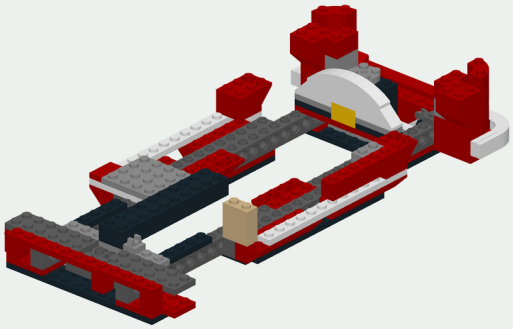
source-10159 | development



Kit 10159; 177 parts; 16 registered
16 reachable movable sites

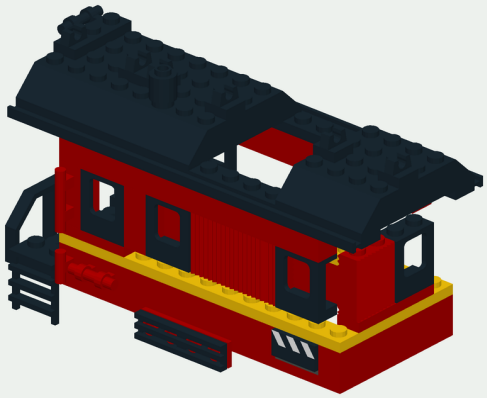
Original-source gallery: assemblies 9–12

source-10220 | development



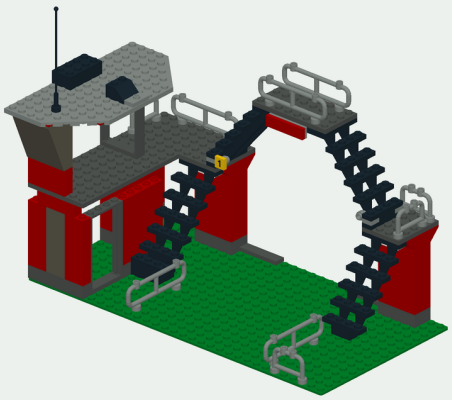
Kit 10220; 153 parts; 16 registered
16 reachable movable sites

source-10014 | heldout-source



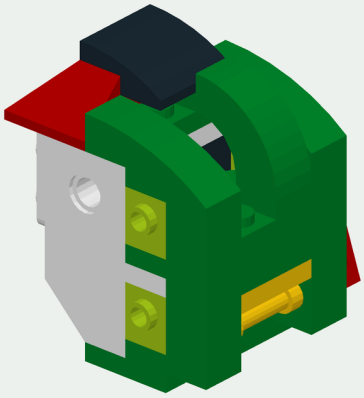
Kit 10014; 71 parts; 16 registered
15 reachable movable sites

source-10128 | heldout-source



Kit 10128; 98 parts; 16 registered
14 reachable movable sites

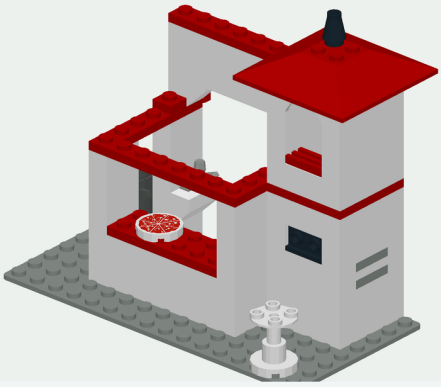
source-omr-31031 | heldout-source



Kit 31031; 42 parts; 12 registered
12 reachable movable sites

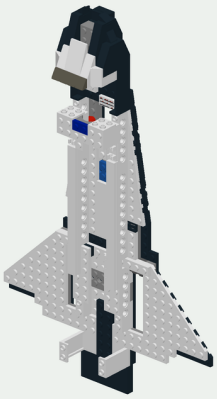
Original-source gallery: assemblies 13–16

source-10036 | heldout-source



Kit 10036; 79 parts; 16 registered
16 reachable movable sites

source-10213 | heldout-source



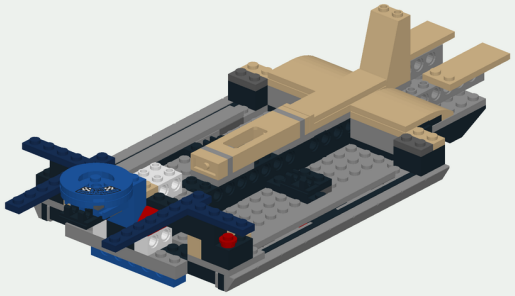
Kit 10213; 179 parts; 2 registered
2 reachable movable sites

source-omr-10242 | heldout-source



Kit 10242; 180 parts; 16 registered
16 reachable movable sites

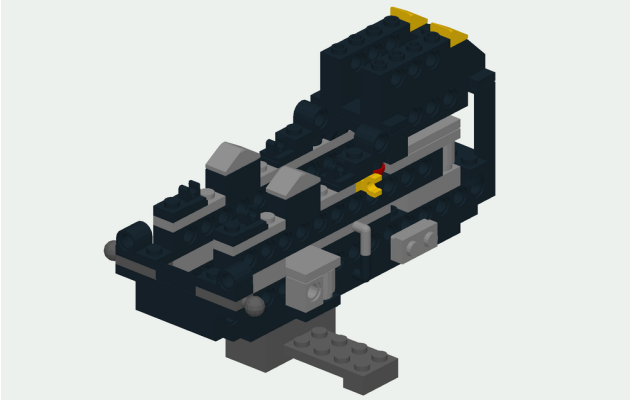
source-omr-10265 | heldout-source



Kit 10265; 179 parts; 4 registered
4 reachable movable sites

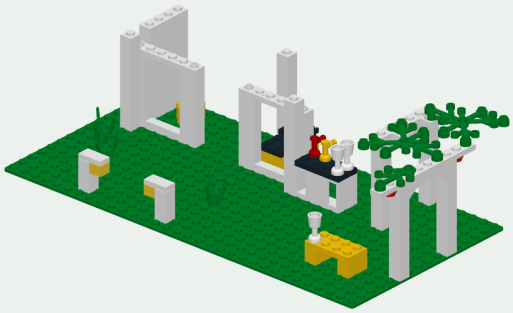
Original-source gallery: assemblies 17–20

source-omr-10269 | heldout-source



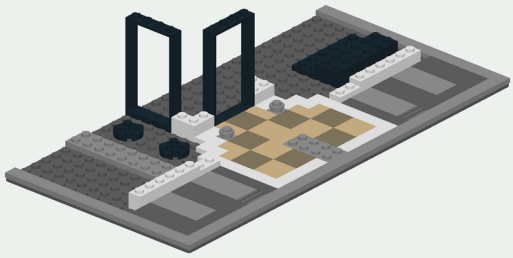
Kit 10269; 105 parts; 16 registered
11 reachable movable sites

source-v23-10037 | heldout-source



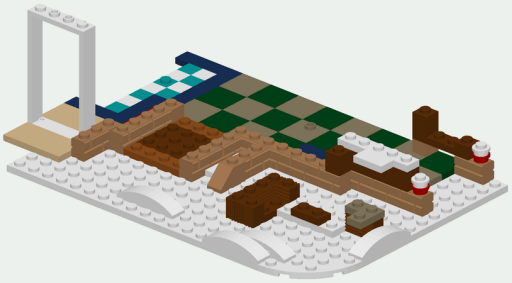
Kit 10037; 65 parts; 16 registered
12 reachable movable sites

source-v23-10184 | heldout-source



Kit 10184; 96 parts; 16 registered
16 reachable movable sites

source-v23-10267 | heldout-source



Kit 10267; 96 parts; 16 registered
16 reachable movable sites

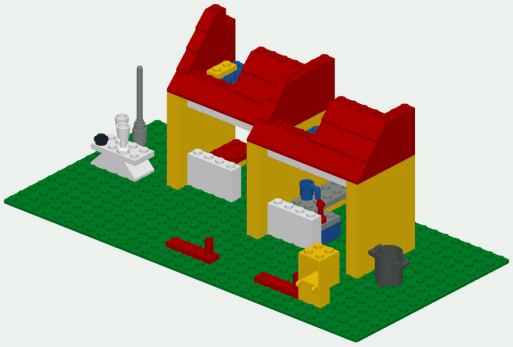
Original-source gallery: assemblies 21–24

source-v23-1472 | heldout-source



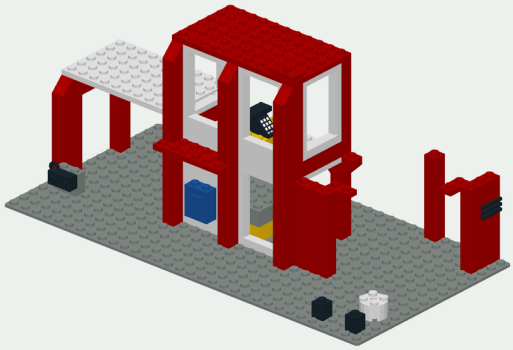
Kit 1472; 89 parts; 16 registered
14 reachable movable sites

source-v23-1484 | heldout-source



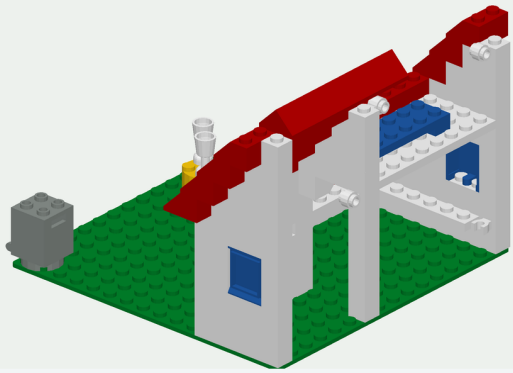
Kit 1484; 96 parts; 16 registered
16 reachable movable sites

source-v23-1490 | heldout-source



Kit 1490; 92 parts; 16 registered
16 reachable movable sites

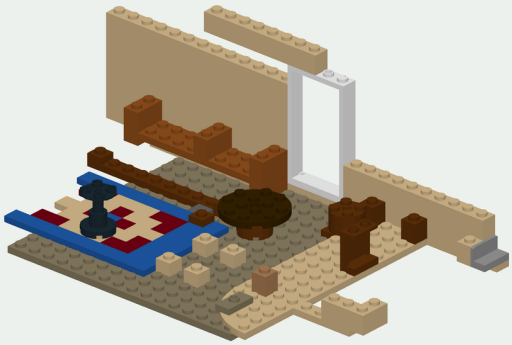
source-v23-1854 | development



Kit 1854; 92 parts; 16 registered
16 reachable movable sites

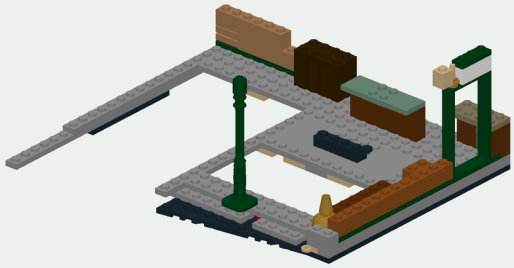
Original-source gallery: assemblies 25–28

source-v23-21302 | heldout-source



Kit 21302; 96 parts; 16 registered
15 reachable movable sites

source-v23-21319 | heldout-source



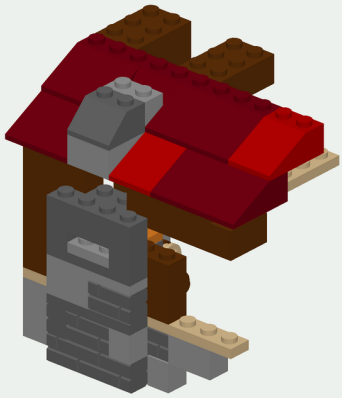
Kit 21319; 96 parts; 16 registered
16 reachable movable sites

source-v23-21325 | heldout-source



Kit 21325; 95 parts; 16 registered
16 reachable movable sites

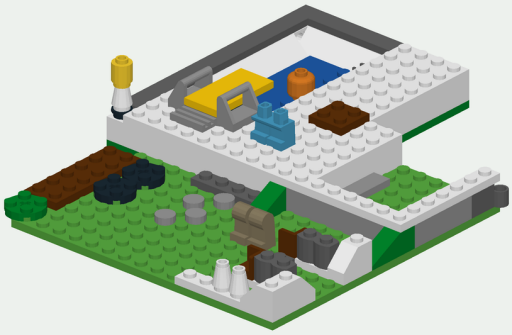
source-v23-31025 | heldout-source



Kit 31025; 95 parts; 16 registered
16 reachable movable sites

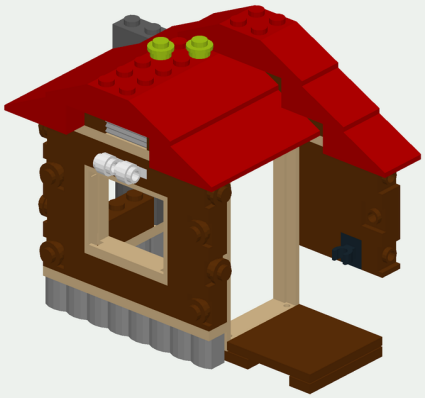
Original-source gallery: assemblies 29–32

source-v23-31038 | development



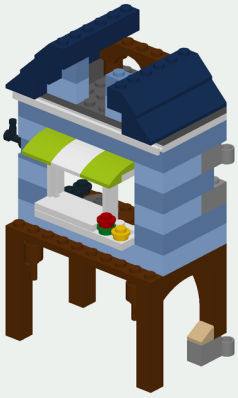
Kit 31038; 94 parts; 16 registered
16 reachable movable sites

source-v23-31048 | development



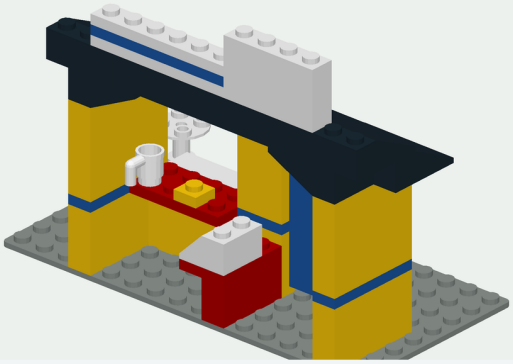
Kit 31048; 96 parts; 16 registered
16 reachable movable sites

source-v23-31063 | development



Kit 31063; 95 parts; 16 registered
16 reachable movable sites

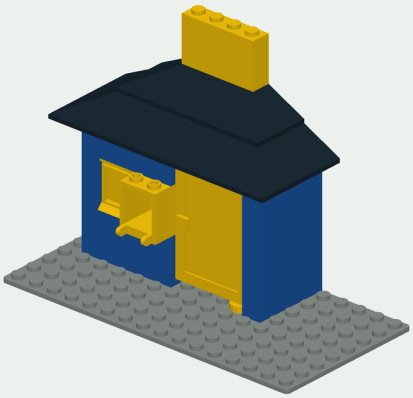
source-v23-6683 | heldout-source



Kit 6683; 50 parts; 16 registered
16 reachable movable sites

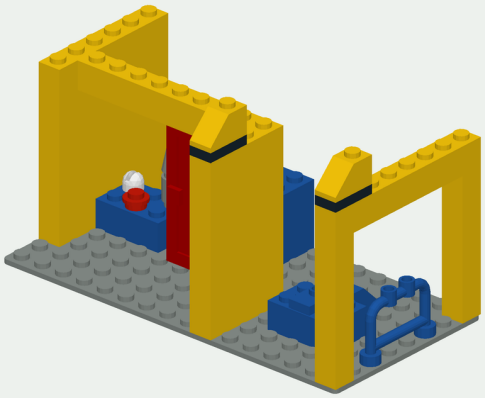
Original-source gallery: assemblies 33–36

source-v23-6689 | heldout-source



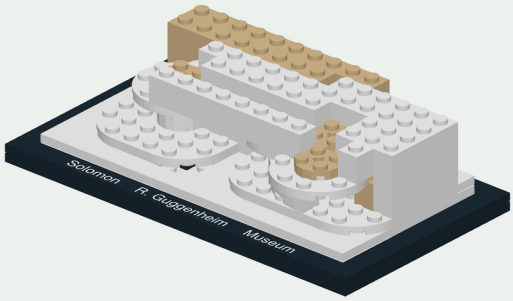
Kit 6689; 39 parts; 16 registered
16 reachable movable sites

source-v23-6699 | heldout-source



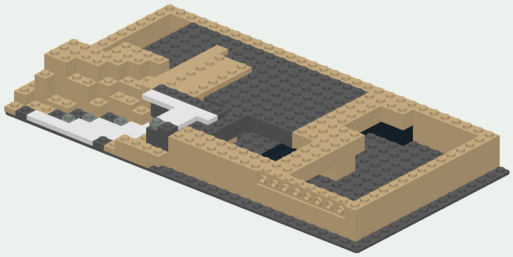
Kit 6699; 44 parts; 16 registered
16 reachable movable sites

source-v23-21004 | heldout-source



Kit 21004; 95 parts; 16 registered
16 reachable movable sites

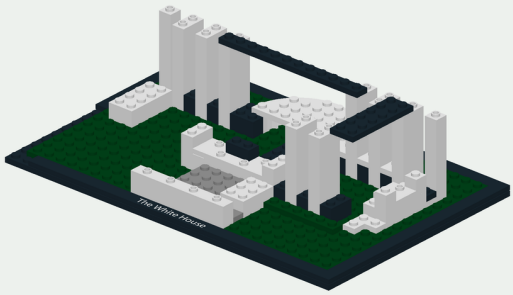
source-v23-21005 | heldout-source



Kit 21005; 96 parts; 16 registered
16 reachable movable sites

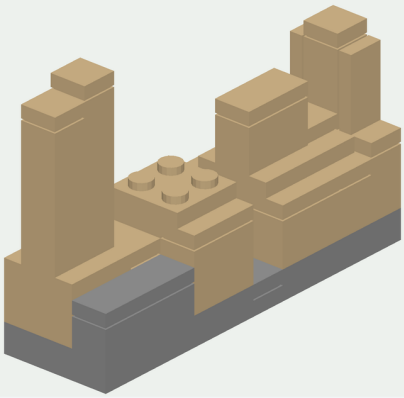
Original-source gallery: assemblies 37–40

source-v23-21006 | development



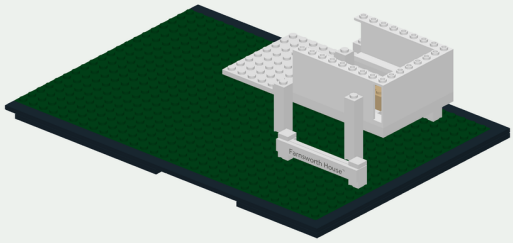
Kit 21006; 96 parts; 16 registered
16 reachable movable sites

source-v23-21007 | development



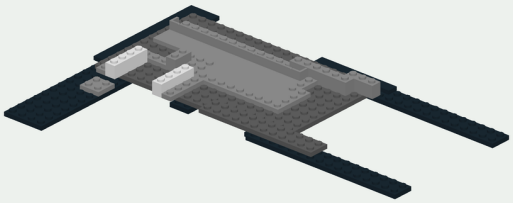
Kit 21007; 46 parts; 16 registered
16 reachable movable sites

source-v23-21009 | heldout-source



Kit 21009; 94 parts; 16 registered
16 reachable movable sites

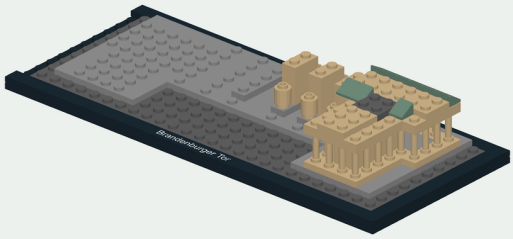
source-v23-21010 | heldout-source



Kit 21010; 87 parts; 16 registered
16 reachable movable sites

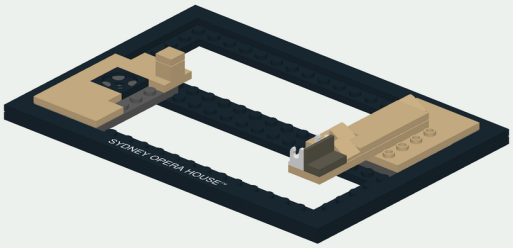
Original-source gallery: assemblies 41–44

source-v23-21011 | development



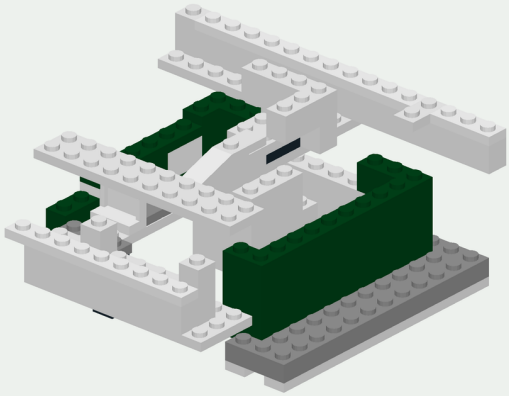
Kit 21011; 91 parts; 16 registered
16 reachable movable sites

source-v23-21012 | development



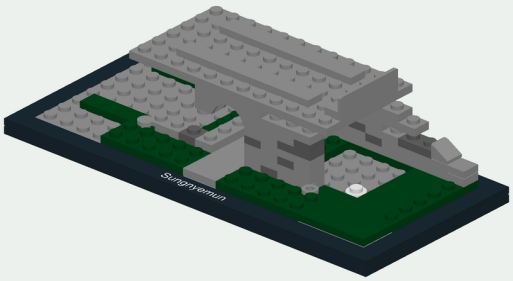
Kit 21012; 48 parts; 16 registered
16 reachable movable sites

source-v23-21014 | heldout-source



Kit 21014; 96 parts; 16 registered
16 reachable movable sites

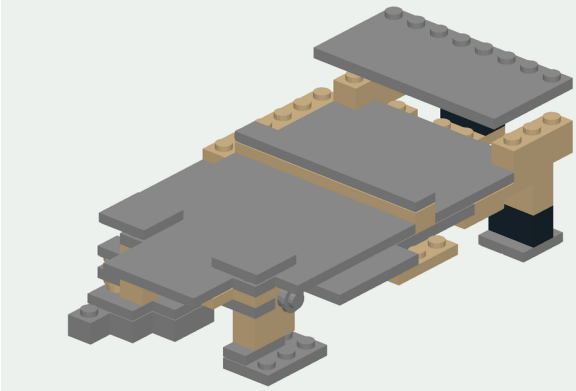
source-v23-21016 | heldout-source



Kit 21016; 95 parts; 16 registered
16 reachable movable sites

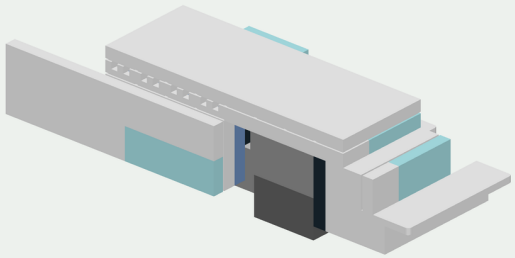
Original-source gallery: assemblies 45–48

source-v23-21017 | heldout-source



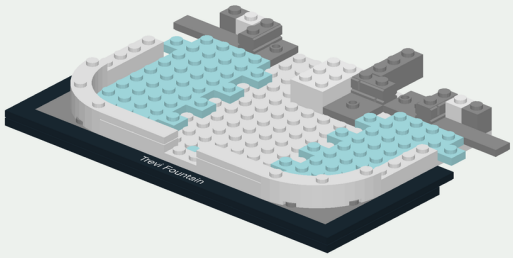
Kit 21017; 96 parts; 16 registered
16 reachable movable sites

source-v23-21018 | development



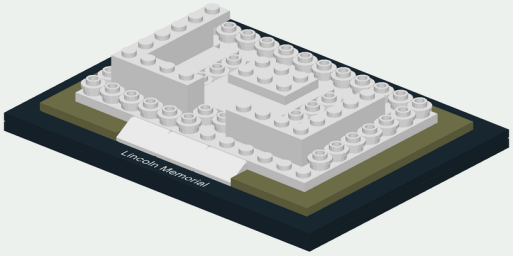
Kit 21018; 60 parts; 16 registered
16 reachable movable sites

source-v23-21020 | heldout-source



Kit 21020; 96 parts; 16 registered
16 reachable movable sites

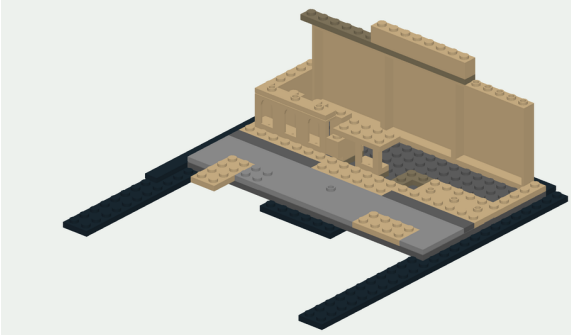
source-v23-21022 | heldout-source



Kit 21022; 96 parts; 16 registered
16 reachable movable sites

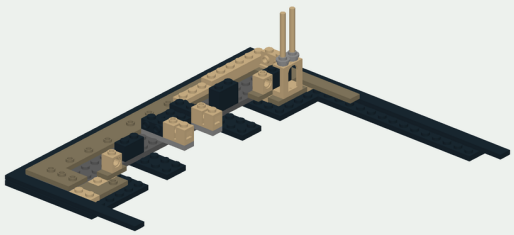
Original-source gallery: assemblies 49–52

source-v23-21024 | heldout-source



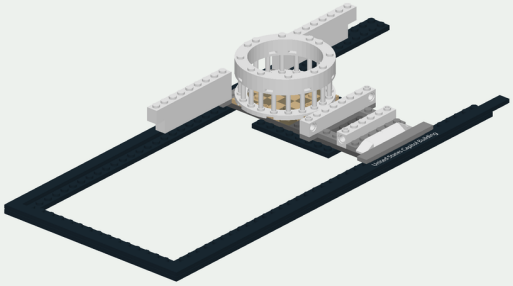
Kit 21024; 96 parts; 16 registered
16 reachable movable sites

source-v23-21029 | heldout-source



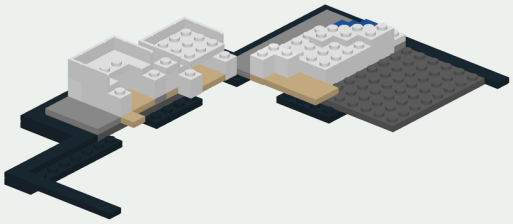
Kit 21029; 76 parts; 16 registered
16 reachable movable sites

source-v23-21030 | heldout-source



Kit 21030; 92 parts; 16 registered
16 reachable movable sites

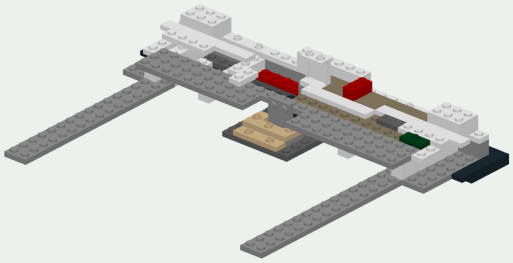
source-v23-21037 | heldout-source



Kit 21037; 96 parts; 16 registered
16 reachable movable sites

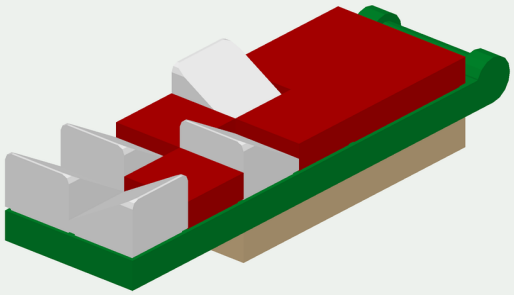
Original-source gallery: assemblies 53–56

source-v23-21045 | development



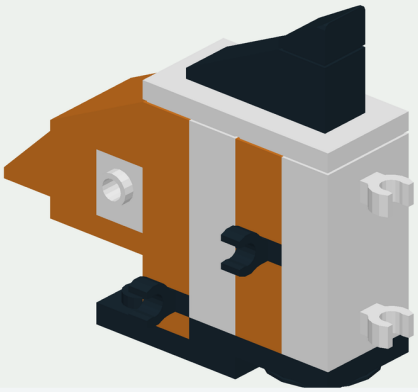
Kit 21045; 83 parts; 16 registered
16 reachable movable sites

source-v23-20015 | development



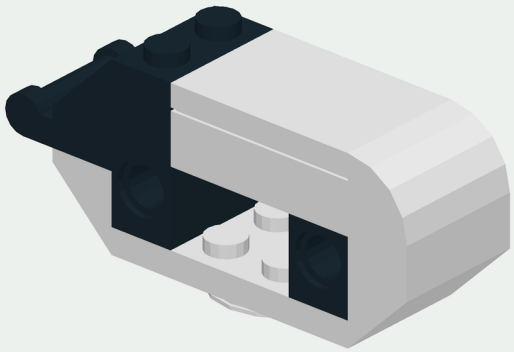
Kit 20015; 13 parts; 12 registered
11 reachable movable sites

source-v23-30025 | heldout-source



Kit 30025; 29 parts; 16 registered
16 reachable movable sites

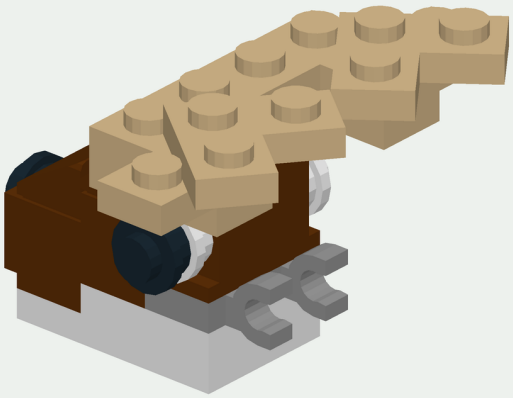
source-v23-30026 | development



Kit 30026; 15 parts; 14 registered
14 reachable movable sites

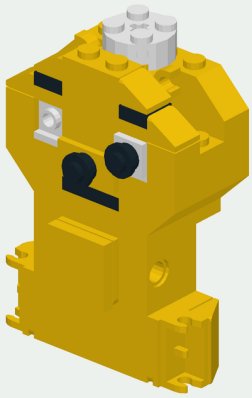
Original-source gallery: assemblies 57–60

source-v23-30027 | development



Kit 30027; 25 parts; 16 registered
15 reachable movable sites

source-v23-30029 | heldout-source



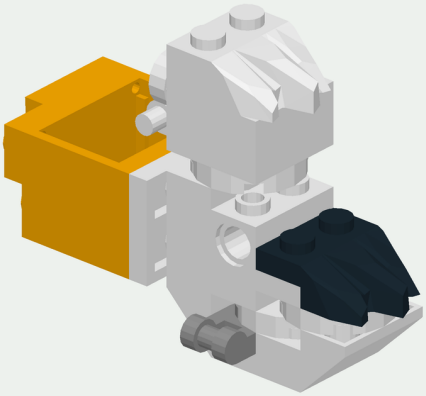
Kit 30029; 64 parts; 16 registered
16 reachable movable sites

source-v23-30542 | heldout-source



Kit 30542; 26 parts; 16 registered
16 reachable movable sites

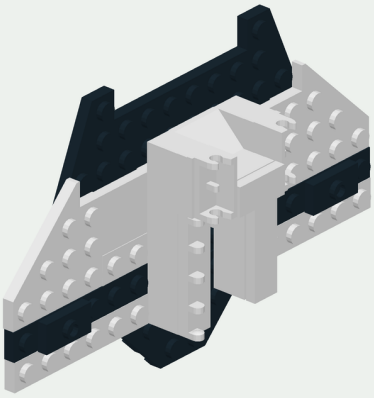
source-v23-30571 | development



Kit 30571; 25 parts; 16 registered
16 reachable movable sites

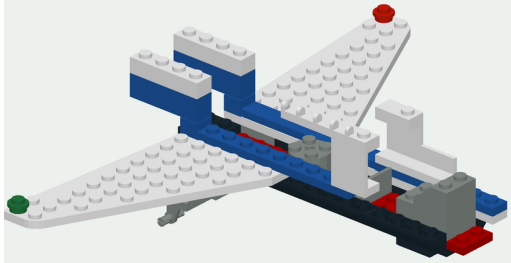
Original-source gallery: assemblies 61–64

source-v23-1682 | heldout-source



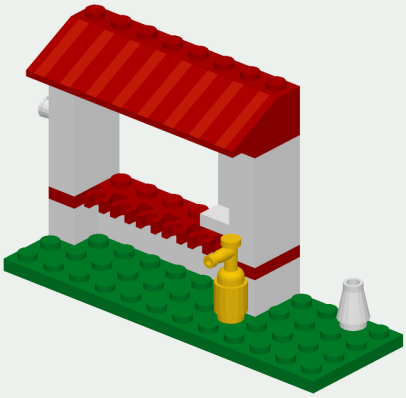
Kit 1682; 30 parts; 12 registered
11 reachable movable sites

source-v23-1775 | heldout-source



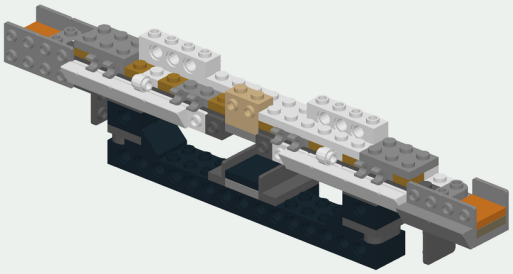
Kit 1775; 44 parts; 15 registered
15 reachable movable sites

source-v23-1808 | heldout-source



Kit 1808; 27 parts; 16 registered
16 reachable movable sites

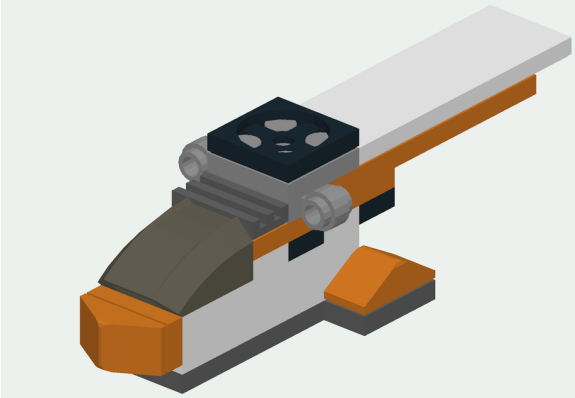
source-v23-21321 | heldout-source



Kit 21321; 91 parts; 16 registered
16 reachable movable sites

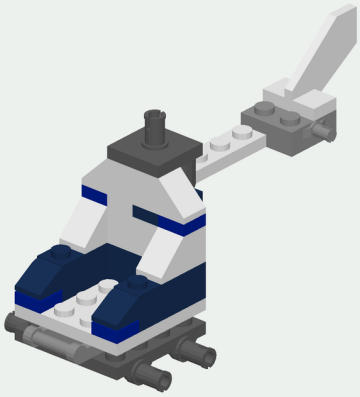
Original-source gallery: assemblies 65–68

source-v23-30181 | heldout-source



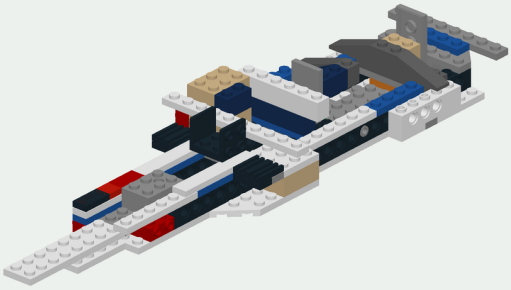
Kit 30181; 27 parts; 8 registered
8 reachable movable sites

source-v23-30311 | heldout-source



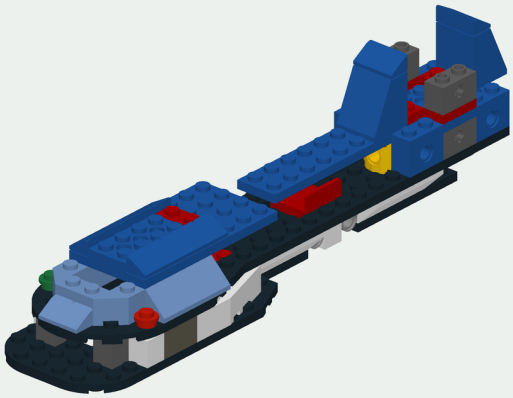
Kit 30311; 27 parts; 16 registered
15 reachable movable sites

source-v23-31039 | development



Kit 31039; 93 parts; 16 registered
16 reachable movable sites

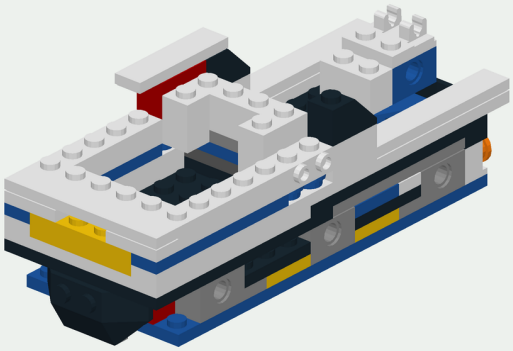
source-v23-31049 | heldout-source



Kit 31049; 87 parts; 15 registered
15 reachable movable sites

Original-source gallery: assemblies 69–72

source-v23-31066 | heldout-source



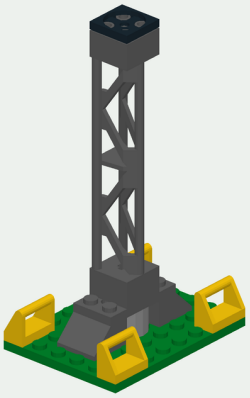
Kit 31066; 93 parts; 16 registered
16 reachable movable sites

source-v23-31092 | development



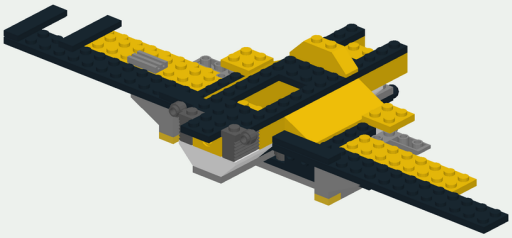
Kit 31092; 60 parts; 16 registered
16 reachable movable sites

source-v23-3181 | development



Kit 3181; 11 parts; 10 registered
10 reachable movable sites

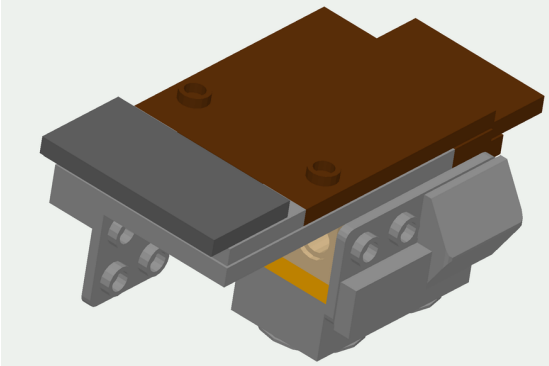
source-v23-6745 | heldout-source



Kit 6745; 94 parts; 16 registered
16 reachable movable sites

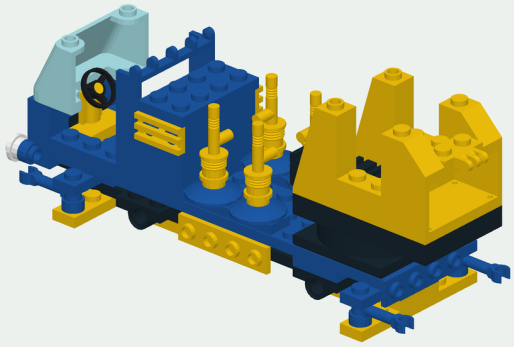
Original-source gallery: assemblies 73–76

source-v23-21303 | heldout-source



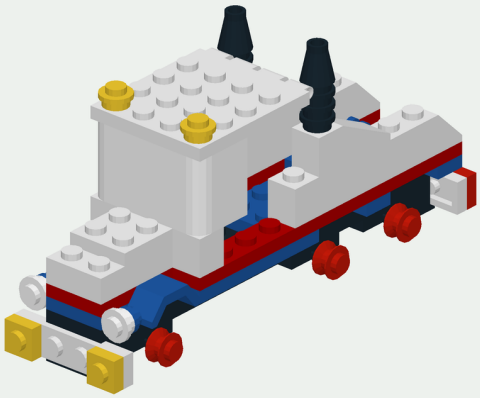
Kit 21303; 33 parts; 14 registered
11 reachable movable sites

source-v23-1489 | heldout-source



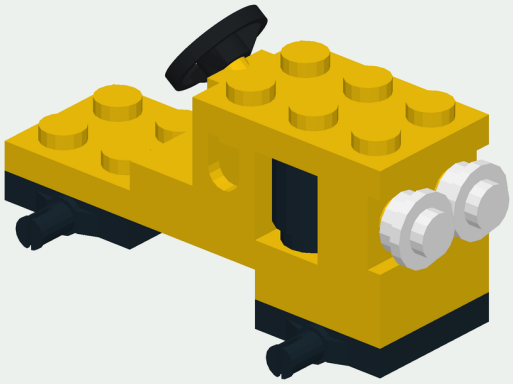
Kit 1489; 93 parts; 8 registered
8 reachable movable sites

source-v23-1572 | development



Kit 1572; 63 parts; 16 registered
16 reachable movable sites

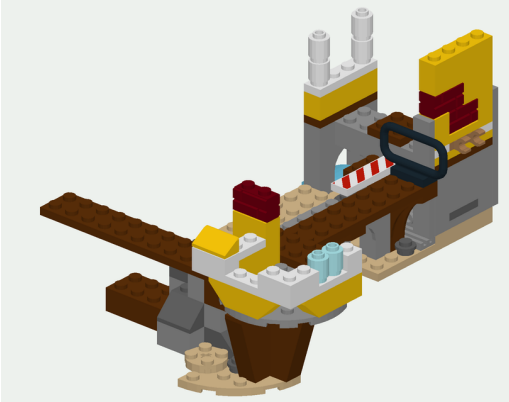
source-v23-1633 | development



Kit 1633; 13 parts; 12 registered
9 reachable movable sites

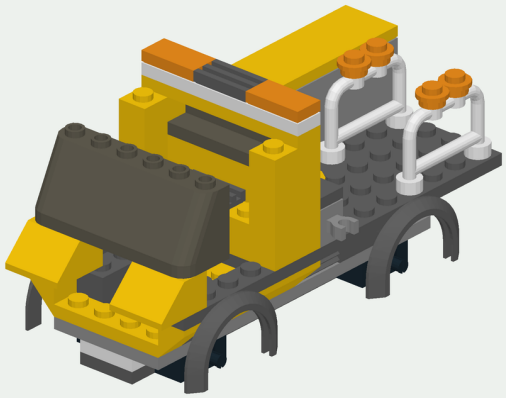
Original-source gallery: assemblies 77–80

source-v23-31084 | heldout-source



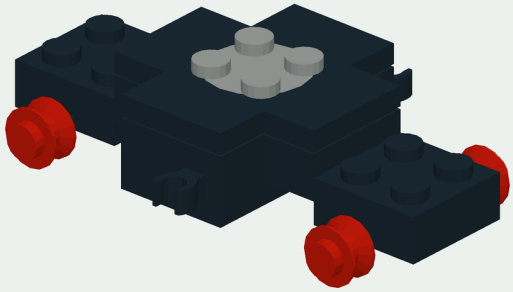
Kit 31084; 96 parts; 12 registered
12 reachable movable sites

source-v23-3179 | development



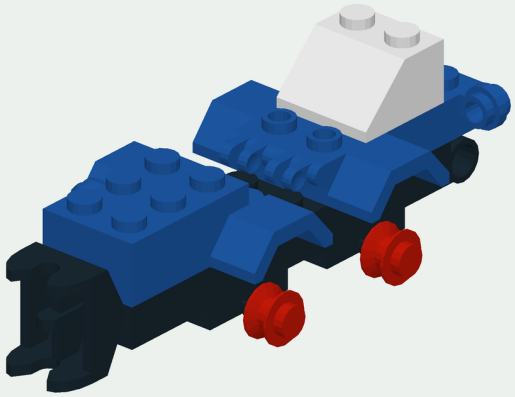
Kit 3179; 60 parts; 16 registered
16 reachable movable sites

source-v23-6678 | development



Kit 6678; 16 parts; 15 registered
12 reachable movable sites

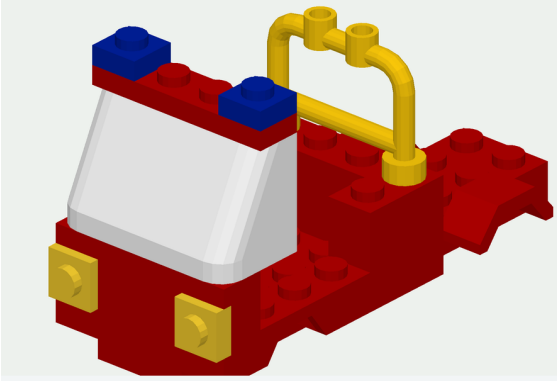
source-v23-6682 | heldout-source



Kit 6682; 20 parts; 16 registered
16 reachable movable sites

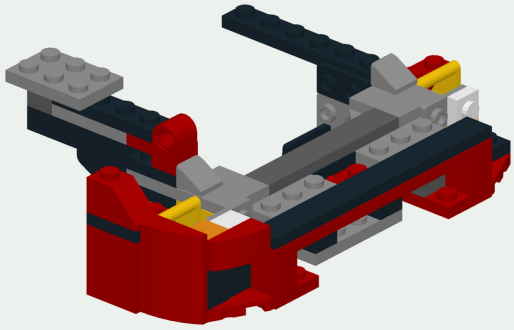
Original-source gallery: assemblies 81–84

source-v23-6690 | heldout-source



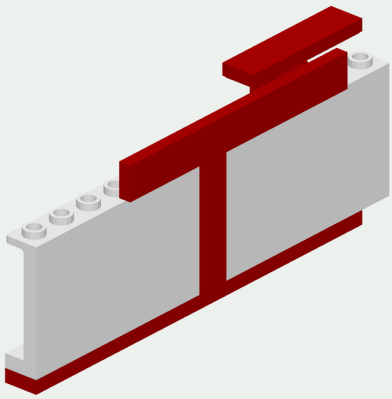
Kit 6690; 25 parts; 16 registered
16 reachable movable sites

source-v23-10248 | heldout-source



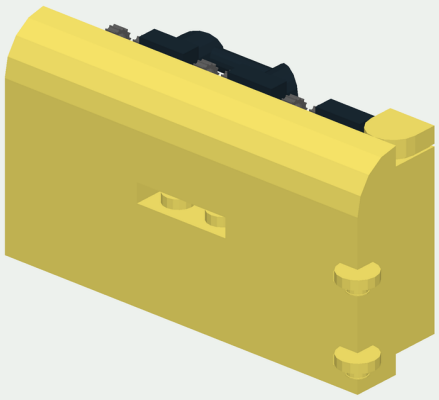
Kit 10248; 92 parts; 12 registered
11 reachable movable sites

source-v23-10258 | heldout-source



Kit 10258; 9 parts; 8 registered
8 reachable movable sites

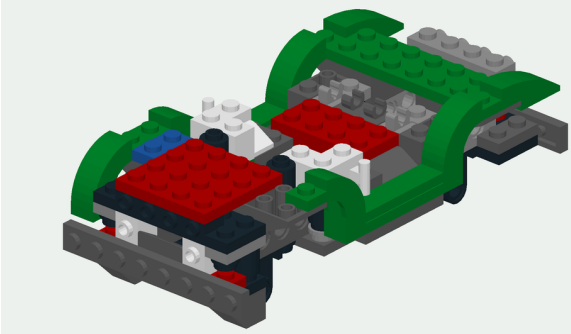
source-v23-10271 | development



Kit 10271; 33 parts; 8 registered
8 reachable movable sites

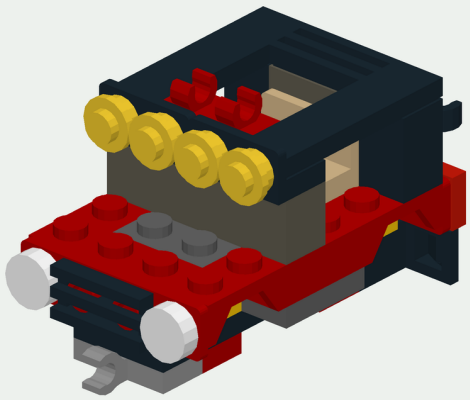
Original-source gallery: assemblies 85–88

source-v23-31037 | heldout-source



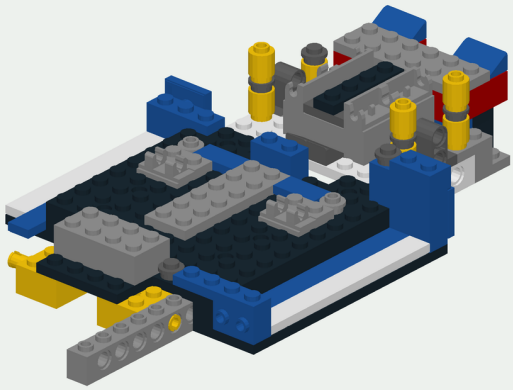
Kit 31037; 96 parts; 16 registered
16 reachable movable sites

source-v23-31040 | development



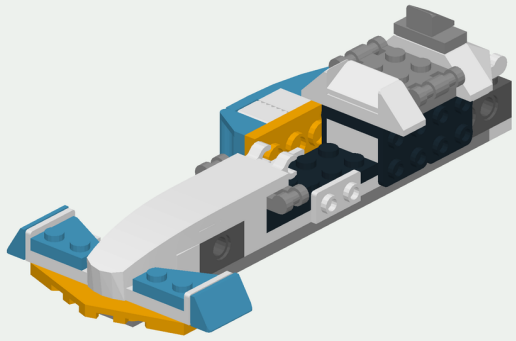
Kit 31040; 49 parts; 16 registered
16 reachable movable sites

source-v23-31070 | development



Kit 31070; 96 parts; 16 registered
16 reachable movable sites

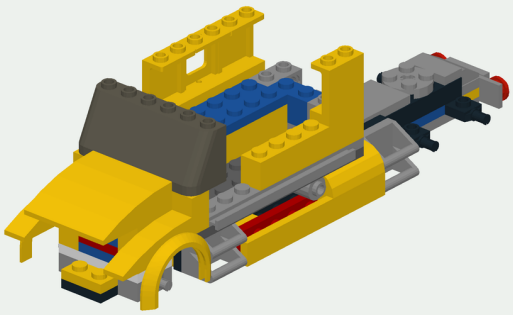
source-v23-31072 | heldout-source



Kit 31072; 56 parts; 12 registered
12 reachable movable sites

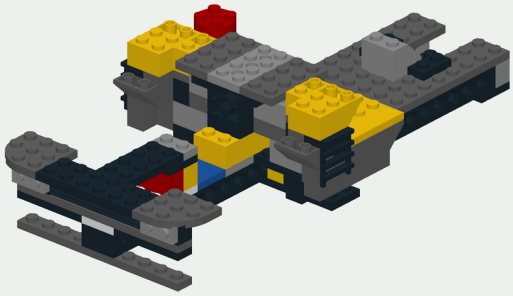
Original-source gallery: assemblies 89–92

source-v23-3221 | heldout-source



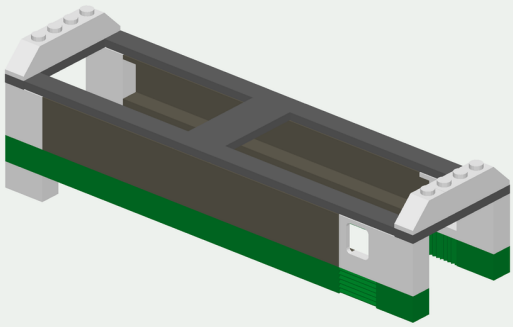
Kit 3221; 91 parts; 16 registered
16 reachable movable sites

source-v23-6753 | heldout-source



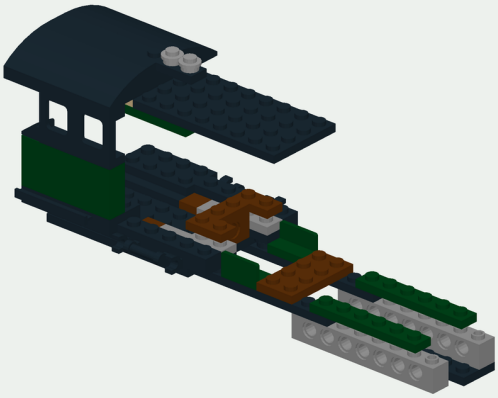
Kit 6753; 96 parts; 16 registered
16 reachable movable sites

source-v23-10158 | development



Kit 10158; 50 parts; 8 registered
8 reachable movable sites

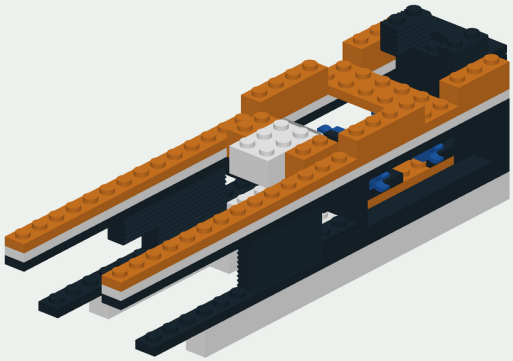
source-v23-10194 | heldout-source



Kit 10194; 90 parts; 12 registered
12 reachable movable sites

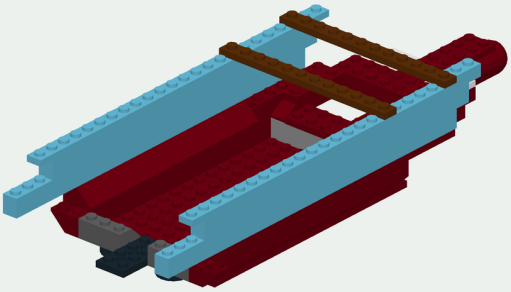
Original-source gallery: assemblies 93–96

source-v23-10233 | heldout-source



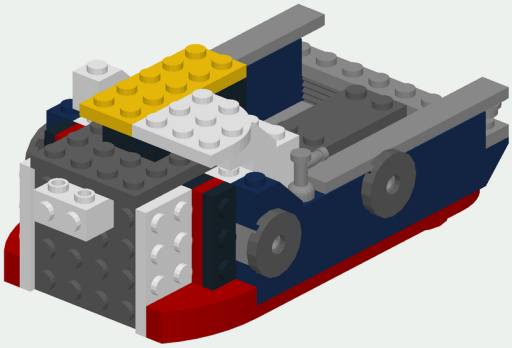
Kit 10233; 96 parts; 16 registered
16 reachable movable sites

source-v23-10241 | heldout-source



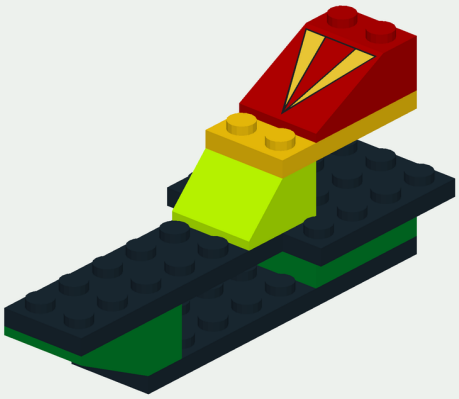
Kit 10241; 96 parts; 16 registered
16 reachable movable sites

source-v23-31045 | heldout-source



Kit 31045; 96 parts; 16 registered
16 reachable movable sites

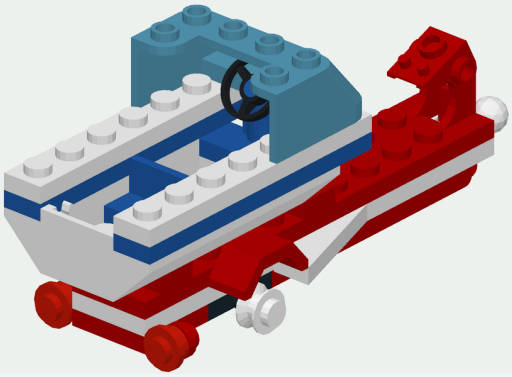
source-v23-4018 | heldout-source



Kit 4018; 9 parts; 8 registered
8 reachable movable sites

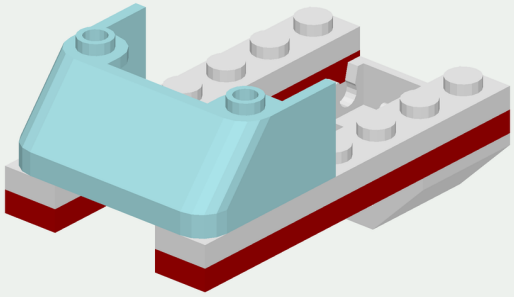
Original-source gallery: assemblies 97–98

source-v23-6698 | heldout-source



Kit 6698; 36 parts; 16 registered
13 reachable movable sites

source-v23-1632 | development



Kit 1632; 11 parts; 8 registered
8 reachable movable sites

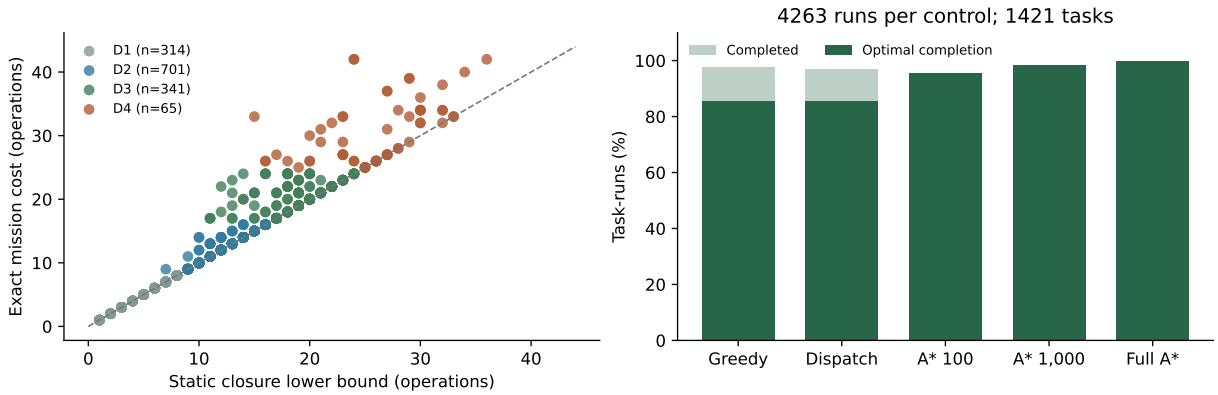
L. Source-bound main-paper cases

Primary inventory-matched pair `source-v23-10037-matched-0/geometry/future-00` from held-out source 10037 in the initial isometric view. Both contexts use 63 source pieces (47 fixed); the exact part-number/color multiset, registered poses, start, target, rack, work, and plan match. After 2 shared legal operations, the exact labels are context 1: first conflict at operation 3; context 2: all legal. Only subsets of original fixed instances change. These publication views omit ID overlays; answers are evaluator annotations. Pixel distinction is not perceptual answerability.

Certified lag-5 repair, `source-10036-35/repair-lag-5`, source 10036. A: the pre-error checkpoint defines the optimal corridor. B: an error and 5 further operations leave it. The last history action is `attach P013`. Undoing it first requires 9 operations to repair; the optimum takes 7, starting with `detach P002`. C: a nonterminal corridor state is restored. These are oracle replay frames, not model reasoning or a model success; exact source instances and actions are archived.

The caption file is generated from the same in-memory case records as the images, and its hash is stored in `figures/provenance.json`. Each geometry member records its exact question and task IDs, source group, profile, start, goal, capacity, jobs, private and public proposed actions, exact label, and full retained inventory. Original source instance IDs, matrices, manifest and mesh hashes, and removed-instance sets bind the illustration to the source geometry. Repair provenance adds pre-error, delayed, and restored states, executed history, shortest repair actions, and inverse-last excess. These are evaluator annotations; they are not model inputs or population results.

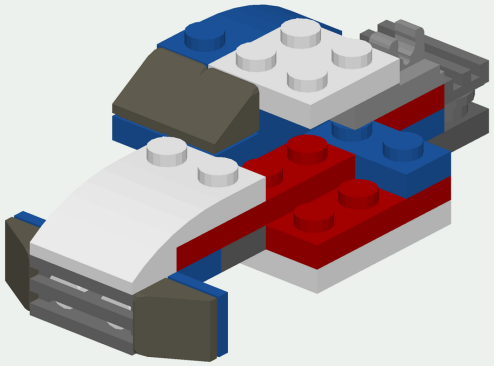
M. Structural control distributions



Exact costs and static lower bounds are shown with overlapping points. Completed and optimal task-runs have different denominators from unique sources. Length bins do not imply monotonic search difficulty. The plot combines sampling cohorts; the release retains separate split and source aggregates, and no timing is a calibrated cross-platform comparison.

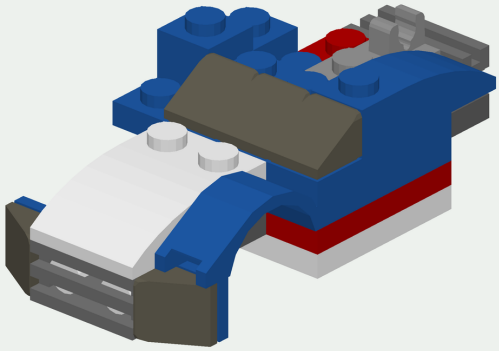
N. A certified optimal resource replay

Operation 00



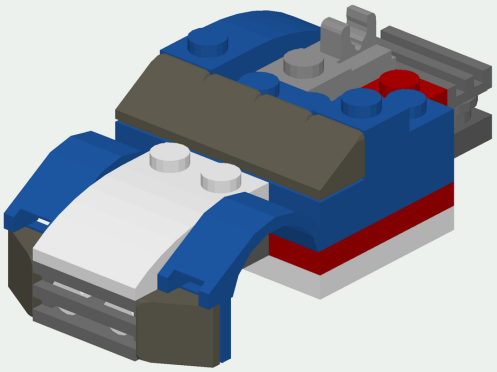
Rack 9/9; remaining cost 42
8 registered pieces installed

Operation 14



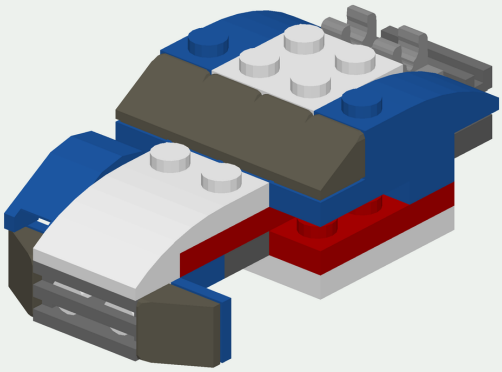
Rack 5/9; remaining cost 28
12 registered pieces installed

Operation 28



Rack 3/9; remaining cost 14
14 registered pieces installed

Operation 42



Rack 1/9; remaining cost 0
16 registered pieces installed

Exact oracle replay of `roadster-body-05` from source 31027. The displayed operations are 0, 14, 28, 42; the certified optimum is 42 operations. This is a software witness, not a model trajectory.All displayed states preserve source poses and certified connectivity. The figure provenance contains the task ID, seed, occupancy, ledger, step indices, and exact remaining cost. Rack holders and off-assembly work are explicit abstractions of the task.

O. Complete episode census

"Jobs" is the declared inspection count, "Done" its initially completed subset, and A* expansions use seed 0 with the closure bound. D1–D4 are length bins. Internal IDs are evaluator documentation; participant IDs are consistently permuted by seed. Exact starts, goals, jobs, witnesses, and oracle paths are available in the task bank.

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
roadster-body-01	D3	17	0	0	17	17	17
roadster-body-02	D4	17	0	0	32	32	32
roadster-body-03	D4	9	0	0	27	23	33
roadster-body-04	D4	9	0	0	27	23	38
roadster-body-05	D4	9	0	0	42	24	2,555
roadster-body-06	D4	9	0	0	42	24	2,555
roadster-body-07	D4	9	2	0	37	27	2,185
roadster-body-08	D4	9	2	0	37	27	2,139
roadster-body-09	D3	9	2	1	20	20	21
roadster-body-10	D3	9	2	1	18	18	18
roadster-body-11	D3	9	1	0	18	18	18
roadster-body-12	D3	9	1	0	18	18	18
roadster-body-13	D4	9	4	0	39	29	2,131
roadster-body-14	D4	9	4	0	39	29	2,129
kart-deck-01	D3	18	0	0	18	18	18
kart-deck-02	D4	18	0	0	25	25	25
kart-deck-03	D3	5	0	0	17	11	32
kart-deck-04	D3	5	0	0	17	11	32
kart-deck-05	D4	5	0	0	26	18	119
kart-deck-06	D3	5	0	0	24	16	90
kart-deck-07	D4	5	2	0	31	21	121
kart-deck-08	D4	5	2	0	29	21	218
kart-deck-09	D3	5	2	1	21	15	68
kart-deck-10	D3	5	2	1	23	19	42
kart-deck-11	D3	5	1	0	23	19	42
kart-deck-12	D3	5	1	0	21	15	68
kart-deck-13	D4	5	4	0	33	23	403
kart-deck-14	D4	5	4	0	33	23	392
airframe-01	D3	21	0	0	21	21	21
airframe-02	D4	21	0	0	33	33	33
airframe-03	D2	8	0	0	16	14	16
airframe-04	D2	8	0	0	16	14	16
airframe-05	D3	8	0	0	17	13	30
airframe-06	D3	8	0	0	17	13	30
airframe-07	D3	8	2	0	22	20	22
airframe-08	D3	8	2	0	22	20	22
airframe-09	D2	8	2	1	9	9	9
airframe-10	D2	8	2	1	9	9	9
airframe-11	D2	8	1	0	9	9	9
airframe-12	D2	8	1	0	9	9	9
airframe-13	D4	8	4	0	26	24	26
airframe-14	D4	8	4	0	26	24	26
truck-cab-01	D3	18	0	0	18	18	18
truck-cab-02	D4	18	0	0	27	27	27
truck-cab-03	D4	12	2	0	40	34	1,522
truck-cab-04	D4	12	2	0	36	30	809
truck-cab-05	D3	12	2	1	21	19	22
truck-cab-06	D3	12	2	1	21	19	22
truck-cab-07	D3	12	1	0	21	19	22
truck-cab-08	D3	12	1	0	21	19	22

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
truck-cab-09	D4	12	4	0	42	36	1,546
truck-cab-10	D4	12	4	0	38	32	831
house-roof-01	D3	18	0	0	18	18	18
house-roof-02	D3	18	0	0	22	22	22
house-roof-03	D3	5	0	0	21	17	37
house-roof-04	D3	5	0	0	21	17	37
house-roof-05	D3	5	0	0	22	18	45
house-roof-06	D3	5	0	0	22	18	45
house-roof-07	D4	5	2	0	27	23	54
house-roof-08	D4	5	2	0	27	23	50
house-roof-09	D3	5	2	1	17	17	19
house-roof-10	D3	5	2	1	17	17	23
house-roof-11	D3	5	1	0	17	17	23
house-roof-12	D3	5	1	0	17	17	19
house-roof-13	D4	5	4	0	33	29	76
house-roof-14	D4	5	4	0	31	27	68
sports-body-01	D2	22	0	0	10	10	10
sports-body-02	D2	22	0	0	11	11	11
sports-body-03	D3	8	2	0	17	17	17
source-10001-01	D1	2	0	0	2	2	2
source-10001-02	D1	2	0	0	1	1	1
source-10001-03	D1	2	2	0	5	5	5
source-10001-04	D1	2	2	0	6	6	6
source-10001-05	D1	2	0	0	2	2	2
source-10001-06	D1	2	0	0	1	1	1
source-10001-07	D1	2	2	0	4	4	4
source-10001-08	D1	2	2	0	4	4	4
source-10159-01	D2	10	0	0	11	11	11
source-10159-02	D2	10	0	0	9	9	9
source-10159-03	D2	10	2	0	12	12	12
source-10159-04	D2	10	2	0	11	11	11
source-10159-05	D3	10	4	0	21	21	21
source-10159-06	D2	10	4	0	14	14	14
source-10159-07	D2	10	0	0	10	10	10
source-10159-08	D2	10	0	0	14	12	45
source-10159-09	D2	10	2	0	14	14	14
source-10159-10	D2	10	2	0	9	9	9
source-10159-11	D2	10	4	0	12	12	12
source-10159-12	D2	10	4	0	16	16	16
source-10220-01	D1	10	0	0	8	8	8
source-10220-02	D1	10	0	0	7	7	7
source-10220-03	D2	10	2	0	14	14	14
source-10220-04	D2	10	2	0	13	13	13
source-10220-05	D2	10	4	0	14	14	14
source-10220-06	D3	10	4	0	19	19	19
source-10220-07	D1	10	0	0	7	7	7
source-10220-08	D2	10	0	0	11	11	11
source-10220-09	D2	10	2	0	12	12	12
source-10220-10	D2	10	2	0	10	10	10
source-10220-11	D2	10	4	0	14	14	14
source-10220-12	D2	10	4	0	14	14	14
source-10014-01	D2	10	0	0	10	10	10
source-10014-02	D3	10	0	0	21	21	22
source-10014-03	D2	10	2	0	10	10	10
source-10014-04	D3	10	2	0	20	20	20
source-10014-05	D4	10	4	0	26	26	29

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-10014-06	D3	10	4	0	21	21	21
source-10014-07	D1	10	0	0	8	8	8
source-10014-08	D2	10	0	0	15	15	15
source-10014-09	D2	10	2	0	14	12	58
source-10014-10	D2	10	2	0	14	14	15
source-10014-11	D3	10	4	0	21	17	4,372
source-10014-12	D2	10	4	0	15	13	52
source-10128-01	D1	10	0	0	5	5	5
source-10128-02	D2	10	0	0	14	14	14
source-10128-03	D2	10	2	0	15	15	15
source-10128-04	D3	10	2	0	18	18	18
source-10128-05	D3	10	4	0	24	24	24
source-10128-06	D3	10	4	0	18	18	18
source-10128-07	D2	10	0	0	10	10	10
source-10128-08	D1	10	0	0	8	8	8
source-10128-09	D2	10	2	0	11	11	11
source-10128-10	D2	10	2	0	14	14	22
source-10128-11	D2	10	4	0	15	15	15
source-10128-12	D3	10	4	0	19	19	19
source-omr-31031-01	D2	8	0	0	12	12	12
source-omr-31031-02	D1	8	0	0	6	6	6
source-omr-31031-03	D2	8	2	0	13	13	13
source-omr-31031-04	D2	8	2	0	11	11	11
source-omr-31031-05	D3	8	4	0	17	17	17
source-omr-31031-06	D3	8	4	0	20	20	20
source-omr-31031-07	D2	8	0	0	13	11	26
source-omr-31031-08	D2	8	0	0	15	15	23
source-omr-31031-09	D1	8	2	0	8	8	8
source-omr-31031-10	D2	8	2	0	10	10	10
source-omr-31031-11	D2	8	4	0	14	14	14
source-omr-31031-12	D2	8	4	0	15	15	15
roadster-body-context-0-01	D4	9	1	0	34	32	954
roadster-body-context-1-01	D4	9	1	0	34	32	953
roadster-body-context-0-02	D4	9	1	0	34	30	1,018
roadster-body-context-1-02	D4	9	1	0	34	30	1,017
roadster-body-context-0-03	D4	9	1	0	34	30	847
roadster-body-context-1-03	D4	9	1	0	34	30	846
roadster-body-context-0-04	D4	9	1	0	32	30	414
roadster-body-context-1-04	D4	9	1	0	32	30	413
source-10036-01	D2	10	0	0	16	16	16
source-10036-02	D1	10	0	0	8	8	8
source-10036-03	D2	10	0	0	12	12	12
source-10036-04	D4	10	0	0	29	23	406
source-10036-05	D2	10	2	0	16	16	16
source-10036-06	D3	10	2	0	17	17	17
source-10036-07	D3	10	2	0	17	17	17
source-10036-08	D2	10	2	0	15	15	15
source-10036-09	D4	10	4	0	27	27	27
source-10036-10	D3	10	4	0	24	24	24
source-10036-11	D4	10	4	0	26	26	26
source-10036-12	D4	10	4	0	34	28	7,595
source-10036-13	D2	10	0	0	11	11	11
source-10036-14	D2	10	0	0	10	10	10
source-10036-15	D2	10	0	0	15	15	27
source-10036-16	D2	10	0	0	12	12	12
source-10036-17	D2	10	2	0	16	16	22

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-10036-18	D4	10	2	0	26	16	3,168
source-10036-19	D3	10	2	0	24	14	2,226
source-10036-20	D2	10	2	0	14	14	21
source-10036-21	D4	10	4	0	30	20	12,578
source-10036-22	D3	10	4	0	24	18	1,714
source-10036-23	D3	10	4	0	19	19	27
source-10036-24	D3	10	4	0	20	20	56
source-10036-25	D2	10	0	0	10	10	10
source-10036-26	D1	10	0	0	6	6	6
source-10036-27	D2	10	0	0	16	16	16
source-10036-28	D2	10	0	0	9	9	9
source-10036-29	D2	10	2	0	13	13	13
source-10036-30	D2	10	2	0	13	13	13
source-10036-31	D2	10	2	0	12	12	12
source-10036-32	D2	10	2	0	13	11	14
source-10036-33	D3	10	4	0	20	20	20
source-10036-34	D3	10	4	0	22	22	22
source-10036-35	D3	10	4	0	22	18	134
source-10036-36	D3	10	4	0	22	22	22
source-10036-37	D2	10	0	0	14	12	60
source-10036-38	D4	10	0	0	27	17	786
source-10036-39	D3	10	0	0	20	14	264
source-10036-40	D1	10	0	0	5	5	7
source-10036-41	D4	10	2	0	33	15	4,640
source-10036-42	D3	10	2	0	17	11	282
source-10036-43	D2	10	2	0	14	12	101
source-10036-44	D3	10	2	0	19	17	236
source-10036-45	D3	10	4	0	19	15	726
source-10036-46	D2	10	4	0	15	13	151
source-10036-47	D4	10	4	0	32	22	11,933
source-10036-48	D2	10	4	0	14	10	311
source-10213-01	D1	2	0	0	2	2	2
source-10213-02	D1	2	0	0	1	1	1
source-10213-03	D1	2	0	0	1	1	1
source-10213-04	D1	2	2	0	6	6	6
source-10213-05	D1	2	2	0	4	4	4
source-10213-06	D1	2	2	0	5	5	5
source-10213-07	D1	2	2	0	5	5	5
source-10213-08	D1	2	0	0	2	2	2
source-10213-09	D1	2	0	0	1	1	1
source-10213-10	D1	2	0	0	1	1	1
source-10213-11	D1	2	2	0	4	4	4
source-10213-12	D1	2	2	0	2	2	2
source-10213-13	D1	2	2	0	3	3	3
source-10213-14	D1	2	2	0	3	3	3
source-10213-15	D1	2	0	0	1	1	1
source-10213-16	D1	2	0	0	2	2	2
source-10213-17	D1	2	0	0	1	1	1
source-10213-18	D1	2	2	0	4	4	4
source-10213-19	D1	2	2	0	5	5	5
source-10213-20	D1	2	2	0	4	4	4
source-10213-21	D1	2	2	0	3	3	3
source-10213-22	D1	2	0	0	1	1	1
source-10213-23	D1	2	0	0	1	1	1
source-10213-24	D1	2	0	0	2	2	2
source-10213-25	D1	2	2	0	3	3	3

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-10213-26	D1	2	2	0	4	4	4
source-10213-27	D1	2	2	0	5	5	5
source-10213-28	D1	2	2	0	4	4	4
source-omr-10242-01	D1	10	0	0	8	8	8
source-omr-10242-02	D2	10	0	0	14	14	14
source-omr-10242-03	D1	10	0	0	7	7	7
source-omr-10242-04	D2	10	0	0	11	11	11
source-omr-10242-05	D2	10	2	0	13	13	13
source-omr-10242-06	D2	10	2	0	16	16	16
source-omr-10242-07	D2	10	2	0	11	11	11
source-omr-10242-08	D2	10	2	0	15	15	15
source-omr-10242-09	D3	10	4	0	17	17	17
source-omr-10242-10	D2	10	4	0	16	16	16
source-omr-10242-11	D2	10	4	0	14	14	14
source-omr-10242-12	D3	10	4	0	17	17	17
source-omr-10242-13	D2	10	0	0	9	9	9
source-omr-10242-14	D1	10	0	0	7	7	7
source-omr-10242-15	D2	10	0	0	9	9	9
source-omr-10242-16	D1	10	0	0	8	8	8
source-omr-10242-17	D2	10	2	0	13	13	13
source-omr-10242-18	D2	10	2	0	16	16	16
source-omr-10242-19	D2	10	2	0	12	12	12
source-omr-10242-20	D2	10	2	0	12	12	12
source-omr-10242-21	D2	10	4	0	14	14	14
source-omr-10242-22	D2	10	4	0	13	13	13
source-omr-10242-23	D2	10	4	0	13	13	13
source-omr-10242-24	D2	10	4	0	15	15	15
source-omr-10242-25	D2	10	0	0	11	11	11
source-omr-10242-26	D1	10	0	0	7	7	7
source-omr-10242-27	D1	10	0	0	6	6	6
source-omr-10242-28	D2	10	0	0	10	10	10
source-omr-10242-29	D2	10	2	0	9	9	9
source-omr-10242-30	D3	10	2	0	17	17	17
source-omr-10242-31	D2	10	2	0	10	10	10
source-omr-10242-32	D2	10	2	0	12	12	12
source-omr-10242-33	D2	10	4	0	14	14	14
source-omr-10242-34	D2	10	4	0	14	14	14
source-omr-10242-35	D2	10	4	0	15	15	15
source-omr-10242-36	D2	10	4	0	14	14	14
source-omr-10242-37	D2	10	0	0	10	10	10
source-omr-10242-38	D1	10	0	0	8	8	8
source-omr-10242-39	D1	10	0	0	7	7	7
source-omr-10242-40	D1	10	0	0	8	8	8
source-omr-10242-41	D2	10	2	0	10	10	10
source-omr-10242-42	D2	10	2	0	11	11	11
source-omr-10242-43	D2	10	2	0	11	11	11
source-omr-10242-44	D2	10	2	0	12	12	12
source-omr-10242-45	D2	10	4	0	16	16	16
source-omr-10242-46	D3	10	4	0	17	17	17
source-omr-10242-47	D3	10	4	0	17	17	17
source-omr-10242-48	D3	10	4	0	18	18	18
source-omr-10265-01	D1	3	0	0	2	2	2
source-omr-10265-02	D1	3	0	0	1	1	1
source-omr-10265-03	D1	3	0	0	1	1	1
source-omr-10265-04	D1	3	0	0	3	3	3
source-omr-10265-05	D1	3	2	0	5	5	5

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-omr-10265-06	D1	3	2	0	7	7	7
source-omr-10265-07	D1	3	2	0	4	4	4
source-omr-10265-08	D1	3	2	0	5	5	5
source-omr-10265-09	D2	3	4	0	9	9	9
source-omr-10265-10	D2	3	4	0	10	10	10
source-omr-10265-11	D2	3	4	0	12	12	12
source-omr-10265-12	D2	3	4	0	9	9	9
source-omr-10265-13	D1	3	0	0	2	2	2
source-omr-10265-14	D1	3	0	0	2	2	2
source-omr-10265-15	D1	3	0	0	2	2	2
source-omr-10265-16	D1	3	0	0	1	1	1
source-omr-10265-17	D1	3	2	0	4	4	4
source-omr-10265-18	D1	3	2	0	7	7	7
source-omr-10265-19	D1	3	2	0	5	5	5
source-omr-10265-20	D1	3	2	0	5	5	5
source-omr-10265-21	D2	3	4	0	9	9	9
source-omr-10265-22	D2	3	4	0	9	9	9
source-omr-10265-23	D2	3	4	0	9	9	9
source-omr-10265-24	D2	3	4	0	9	9	9
source-omr-10265-25	D1	3	0	0	2	2	2
source-omr-10265-26	D1	3	0	0	3	3	3
source-omr-10265-27	D1	3	0	0	3	3	3
source-omr-10265-28	D1	3	0	0	2	2	2
source-omr-10265-29	D1	3	2	0	3	3	3
source-omr-10265-30	D1	3	2	0	4	4	4
source-omr-10265-31	D1	3	2	0	5	5	5
source-omr-10265-32	D1	3	2	0	5	5	5
source-omr-10265-33	D1	3	4	0	7	7	7
source-omr-10265-34	D1	3	4	0	7	7	7
source-omr-10265-35	D1	3	4	0	7	7	7
source-omr-10265-36	D1	3	4	0	8	8	8
source-omr-10265-37	D1	3	0	0	2	2	2
source-omr-10265-38	D1	3	0	0	3	3	3
source-omr-10265-39	D1	3	0	0	1	1	1
source-omr-10265-40	D1	3	0	0	1	1	1
source-omr-10265-41	D1	3	2	0	8	8	8
source-omr-10265-42	D1	3	2	0	6	6	6
source-omr-10265-43	D1	3	2	0	7	7	7
source-omr-10265-44	D1	3	2	0	6	6	6
source-omr-10265-45	D2	3	4	0	9	9	9
source-omr-10265-46	D2	3	4	0	9	9	9
source-omr-10265-47	D2	3	4	0	9	9	9
source-omr-10265-48	D1	3	4	0	8	8	8
source-omr-10269-01	D2	10	0	0	11	11	11
source-omr-10269-02	D2	10	0	0	13	13	13
source-omr-10269-03	D2	10	0	0	12	12	12
source-omr-10269-04	D2	10	0	0	9	9	9
source-omr-10269-05	D3	10	2	0	17	17	17
source-omr-10269-06	D3	10	2	0	18	18	18
source-omr-10269-07	D2	10	2	0	16	16	16
source-omr-10269-08	D2	10	2	0	15	15	15
source-omr-10269-09	D3	10	4	0	22	22	22
source-omr-10269-10	D3	10	4	0	21	21	21
source-omr-10269-11	D3	10	4	0	19	19	19
source-omr-10269-12	D3	10	4	0	20	20	20
source-omr-10269-13	D1	10	0	0	6	6	6

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-omr-10269-14	D2	10	0	0	11	11	11
source-omr-10269-15	D2	10	0	0	10	10	10
source-omr-10269-16	D1	10	0	0	7	7	7
source-omr-10269-17	D2	10	2	0	14	14	14
source-omr-10269-18	D2	10	2	0	15	15	15
source-omr-10269-19	D2	10	2	0	11	11	11
source-omr-10269-20	D2	10	2	0	12	12	12
source-omr-10269-21	D3	10	4	0	18	18	18
source-omr-10269-22	D2	10	4	0	16	16	16
source-omr-10269-23	D2	10	4	0	13	13	13
source-omr-10269-24	D2	10	4	0	14	14	14
source-omr-10269-25	D2	10	0	0	14	14	14
source-omr-10269-26	D1	10	0	0	7	7	7
source-omr-10269-27	D2	10	0	0	11	11	11
source-omr-10269-28	D2	10	0	0	10	10	10
source-omr-10269-29	D2	10	2	0	12	12	12
source-omr-10269-30	D2	10	2	0	13	13	13
source-omr-10269-31	D2	10	2	0	14	14	14
source-omr-10269-32	D2	10	2	0	12	12	12
source-omr-10269-33	D2	10	4	0	15	15	15
source-omr-10269-34	D3	10	4	0	17	17	17
source-omr-10269-35	D2	10	4	0	14	14	14
source-omr-10269-36	D2	10	4	0	15	15	15
source-omr-10269-37	D2	10	0	0	12	12	12
source-omr-10269-38	D2	10	0	0	13	13	13
source-omr-10269-39	D2	10	0	0	12	12	12
source-omr-10269-40	D1	10	0	0	8	8	8
source-omr-10269-41	D2	10	2	0	15	15	15
source-omr-10269-42	D2	10	2	0	13	13	13
source-omr-10269-43	D2	10	2	0	12	12	14
source-omr-10269-44	D2	10	2	0	16	16	16
source-omr-10269-45	D2	10	4	0	16	16	16
source-omr-10269-46	D3	10	4	0	18	18	18
source-omr-10269-47	D3	10	4	0	17	17	17
source-omr-10269-48	D2	10	4	0	15	15	15
source-10014-context-0-01	D3	10	1	0	18	18	23
source-10014-context-1-01	D3	10	1	0	18	18	18
source-10014-context-0-02	D3	10	1	0	22	22	31
source-10014-context-1-02	D3	10	1	0	22	22	22
source-10036-context-0-01	D2	10	0	0	15	15	15
source-10036-context-1-01	D2	10	0	0	15	15	15
source-10036-context-0-02	D2	10	0	0	16	16	16
source-10036-context-1-02	D2	10	0	0	16	16	16
source-10036-intervention-2-context-0-01	D3	10	0	0	17	17	17
source-10036-intervention-2-context-1-01	D3	10	0	0	17	17	17
source-10036-intervention-2-context-0-02	D2	10	0	0	15	15	15
source-10036-intervention-2-context-1-02	D2	10	0	0	15	15	15
source-v23-10037-01	D2	10	0	0	9	9	9
source-v23-10037-02	D2	10	0	0	9	9	9
source-v23-10037-03	D2	10	2	0	12	12	12
source-v23-10037-04	D2	10	2	0	12	12	12
source-v23-10037-05	D2	10	4	0	12	12	12
source-v23-10037-06	D3	10	4	0	18	18	18
source-v23-10037-07	D1	10	0	0	6	6	6
source-v23-10037-08	D2	10	0	0	12	12	12
source-v23-10037-09	D2	10	2	0	13	13	13

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-10037-10	D2	10	2	0	11	11	11
source-v23-10037-11	D2	10	4	0	10	10	10
source-v23-10037-12	D2	10	4	0	13	13	13
source-v23-10184-01	D1	10	0	0	7	7	7
source-v23-10184-02	D1	10	0	0	7	7	7
source-v23-10184-03	D2	10	2	0	11	11	11
source-v23-10184-04	D2	10	2	0	13	13	13
source-v23-10184-05	D2	10	4	0	12	12	12
source-v23-10184-06	D3	10	4	0	17	17	17
source-v23-10184-07	D1	10	0	0	5	5	5
source-v23-10184-08	D1	10	0	0	7	7	7
source-v23-10184-09	D2	10	2	0	12	12	12
source-v23-10184-10	D1	10	2	0	7	7	7
source-v23-10184-11	D2	10	4	0	12	12	12
source-v23-10184-12	D2	10	4	0	9	9	9
source-v23-10267-01	D1	10	0	0	5	5	5
source-v23-10267-02	D2	10	0	0	9	9	9
source-v23-10267-03	D2	10	2	0	12	12	12
source-v23-10267-04	D2	10	2	0	15	15	15
source-v23-10267-05	D3	10	4	0	20	20	20
source-v23-10267-06	D3	10	4	0	19	19	19
source-v23-10267-07	D1	10	0	0	8	8	8
source-v23-10267-08	D1	10	0	0	8	8	8
source-v23-10267-09	D2	10	2	0	10	10	10
source-v23-10267-10	D2	10	2	0	13	13	13
source-v23-10267-11	D2	10	4	0	13	13	13
source-v23-10267-12	D2	10	4	0	14	14	14
source-v23-1472-01	D2	10	0	0	10	10	10
source-v23-1472-02	D1	10	0	0	7	7	7
source-v23-1472-03	D2	10	2	0	15	15	15
source-v23-1472-04	D2	10	2	0	13	13	13
source-v23-1472-05	D3	10	4	0	19	19	19
source-v23-1472-06	D3	10	4	0	20	20	20
source-v23-1472-07	D1	10	0	0	5	5	5
source-v23-1472-08	D1	10	0	0	8	8	8
source-v23-1472-09	D2	10	2	0	11	11	11
source-v23-1472-10	D2	10	2	0	12	12	12
source-v23-1472-11	D2	10	4	0	12	12	12
source-v23-1472-12	D2	10	4	0	13	13	13
source-v23-1484-01	D2	10	0	0	12	12	12
source-v23-1484-02	D3	10	0	0	17	17	17
source-v23-1484-03	D2	10	2	0	16	16	16
source-v23-1484-04	D2	10	2	0	16	16	16
source-v23-1484-05	D3	10	4	0	19	19	19
source-v23-1484-06	D2	10	4	0	15	15	15
source-v23-1484-07	D2	10	0	0	10	10	10
source-v23-1484-08	D2	10	0	0	9	9	9
source-v23-1484-09	D3	10	2	0	17	17	17
source-v23-1484-10	D2	10	2	0	14	14	14
source-v23-1484-11	D2	10	4	0	16	16	16
source-v23-1484-12	D2	10	4	0	12	12	12
source-v23-1490-01	D2	10	0	0	10	10	10
source-v23-1490-02	D2	10	0	0	11	11	11
source-v23-1490-03	D2	10	2	0	15	15	15
source-v23-1490-04	D2	10	2	0	15	15	15
source-v23-1490-05	D3	10	4	0	17	17	17

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-1490-06	D2	10	4	0	16	16	16
source-v23-1490-07	D1	10	0	0	5	5	5
source-v23-1490-08	D2	10	0	0	10	10	10
source-v23-1490-09	D2	10	2	0	14	14	14
source-v23-1490-10	D2	10	2	0	12	12	12
source-v23-1490-11	D2	10	4	0	14	14	14
source-v23-1490-12	D2	10	4	0	15	15	15
source-v23-1854-01	D2	10	0	0	9	9	9
source-v23-1854-02	D2	10	0	0	15	15	15
source-v23-1854-03	D3	10	2	0	18	18	18
source-v23-1854-04	D2	10	2	0	11	11	11
source-v23-1854-05	D3	10	4	0	17	17	17
source-v23-1854-06	D3	10	4	0	21	21	21
source-v23-1854-07	D2	10	0	0	13	13	13
source-v23-1854-08	D2	10	0	0	13	11	19
source-v23-1854-09	D3	10	2	0	18	18	18
source-v23-1854-10	D2	10	2	0	16	16	18
source-v23-1854-11	D3	10	4	0	17	17	17
source-v23-1854-12	D3	10	4	0	17	17	17
source-v23-21302-01	D2	10	0	0	15	15	15
source-v23-21302-02	D1	10	0	0	6	6	6
source-v23-21302-03	D2	10	2	0	16	16	16
source-v23-21302-04	D3	10	2	0	22	22	22
source-v23-21302-05	D3	10	4	0	21	21	21
source-v23-21302-06	D3	10	4	0	20	20	20
source-v23-21302-07	D3	10	0	0	17	15	54
source-v23-21302-08	D1	10	0	0	6	6	6
source-v23-21302-09	D2	10	2	0	11	11	11
source-v23-21302-10	D3	10	2	0	19	17	157
source-v23-21302-11	D3	10	4	0	21	19	430
source-v23-21302-12	D2	10	4	0	12	12	12
source-v23-21319-01	D1	10	0	0	8	8	8
source-v23-21319-02	D1	10	0	0	7	7	7
source-v23-21319-03	D3	10	2	0	19	19	19
source-v23-21319-04	D3	10	2	0	18	18	18
source-v23-21319-05	D3	10	4	0	22	22	22
source-v23-21319-06	D3	10	4	0	20	20	20
source-v23-21319-07	D1	10	0	0	8	8	8
source-v23-21319-08	D1	10	0	0	8	8	8
source-v23-21319-09	D3	10	2	0	17	17	25
source-v23-21319-10	D2	10	2	0	13	13	13
source-v23-21319-11	D3	10	4	0	18	18	81
source-v23-21319-12	D2	10	4	0	16	16	31
source-v23-21325-01	D2	10	0	0	16	16	16
source-v23-21325-02	D2	10	0	0	14	14	14
source-v23-21325-03	D2	10	2	0	13	13	13
source-v23-21325-04	D2	10	2	0	15	15	15
source-v23-21325-05	D3	10	4	0	18	18	18
source-v23-21325-06	D3	10	4	0	17	17	17
source-v23-21325-07	D1	10	0	0	8	8	8
source-v23-21325-08	D2	10	0	0	9	9	9
source-v23-21325-09	D2	10	2	0	13	13	13
source-v23-21325-10	D3	10	2	0	17	15	73
source-v23-21325-11	D2	10	4	0	15	15	15
source-v23-21325-12	D2	10	4	0	14	14	14
source-v23-31025-01	D1	10	0	0	8	8	8

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-31025-02	D1	10	0	0	8	8	8
source-v23-31025-03	D2	10	2	0	15	15	15
source-v23-31025-04	D2	10	2	0	16	16	16
source-v23-31025-05	D3	10	4	0	24	24	24
source-v23-31025-06	D3	10	4	0	17	17	17
source-v23-31025-07	D1	10	0	0	4	4	4
source-v23-31025-08	D2	10	0	0	9	9	11
source-v23-31025-09	D2	10	2	0	12	12	12
source-v23-31025-10	D2	10	2	0	13	13	13
source-v23-31025-11	D3	10	4	0	17	17	17
source-v23-31025-12	D2	10	4	0	12	12	14
source-v23-31038-01	D2	10	0	0	12	12	12
source-v23-31038-02	D2	10	0	0	10	10	10
source-v23-31038-03	D2	10	2	0	11	11	11
source-v23-31038-04	D2	10	2	0	13	13	13
source-v23-31038-05	D4	10	4	0	33	33	33
source-v23-31038-06	D4	10	4	0	26	26	26
source-v23-31038-07	D1	10	0	0	6	6	6
source-v23-31038-08	D2	10	0	0	9	9	9
source-v23-31038-09	D2	10	2	0	12	12	12
source-v23-31038-10	D2	10	2	0	12	12	12
source-v23-31038-11	D3	10	4	0	22	22	109
source-v23-31038-12	D3	10	4	0	24	24	43
source-v23-31048-01	D2	10	0	0	12	12	12
source-v23-31048-02	D1	10	0	0	8	8	8
source-v23-31048-03	D3	10	2	0	22	22	22
source-v23-31048-04	D3	10	2	0	23	23	23
source-v23-31048-05	D4	10	4	0	26	26	27
source-v23-31048-06	D4	10	4	0	26	26	26
source-v23-31048-07	D2	10	0	0	12	12	12
source-v23-31048-08	D2	10	0	0	12	12	12
source-v23-31048-09	D2	10	2	0	15	15	15
source-v23-31048-10	D3	10	2	0	17	17	17
source-v23-31048-11	D3	10	4	0	23	23	23
source-v23-31048-12	D3	10	4	0	21	21	21
source-v23-31063-01	D2	10	0	0	10	10	10
source-v23-31063-02	D1	10	0	0	7	7	7
source-v23-31063-03	D2	10	2	0	14	14	14
source-v23-31063-04	D2	10	2	0	14	14	14
source-v23-31063-05	D2	10	4	0	16	16	16
source-v23-31063-06	D2	10	4	0	16	16	16
source-v23-31063-07	D2	10	0	0	9	9	9
source-v23-31063-08	D2	10	0	0	10	10	10
source-v23-31063-09	D2	10	2	0	12	12	12
source-v23-31063-10	D2	10	2	0	13	13	13
source-v23-31063-11	D2	10	4	0	15	15	15
source-v23-31063-12	D2	10	4	0	11	11	11
source-v23-6683-01	D2	10	0	0	12	12	12
source-v23-6683-02	D2	10	0	0	16	16	16
source-v23-6683-03	D3	10	2	0	18	18	18
source-v23-6683-04	D3	10	2	0	21	21	21
source-v23-6683-05	D4	10	4	0	27	27	33
source-v23-6683-06	D4	10	4	0	29	29	30
source-v23-6683-07	D2	10	0	0	13	13	18
source-v23-6683-08	D2	10	0	0	10	10	12
source-v23-6683-09	D3	10	2	0	24	16	122

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-6683-10	D3	10	2	0	19	13	260
source-v23-6683-11	D3	10	4	0	17	17	18
source-v23-6683-12	D3	10	4	0	17	17	39
source-v23-6689-01	D2	10	0	0	10	10	10
source-v23-6689-02	D3	10	0	0	18	18	18
source-v23-6689-03	D2	10	2	0	16	16	16
source-v23-6689-04	D2	10	2	0	15	15	15
source-v23-6689-05	D3	10	4	0	21	21	21
source-v23-6689-06	D4	10	4	0	25	25	26
source-v23-6689-07	D2	10	0	0	12	12	12
source-v23-6689-08	D2	10	0	0	13	13	13
source-v23-6689-09	D2	10	2	0	15	15	15
source-v23-6689-10	D2	10	2	0	14	14	14
source-v23-6689-11	D3	10	4	0	20	14	446
source-v23-6689-12	D3	10	4	0	20	20	26
source-v23-6699-01	D2	10	0	0	14	14	14
source-v23-6699-02	D2	10	0	0	14	14	14
source-v23-6699-03	D2	10	2	0	16	16	16
source-v23-6699-04	D2	10	2	0	14	14	14
source-v23-6699-05	D2	10	4	0	14	14	14
source-v23-6699-06	D2	10	4	0	16	16	16
source-v23-6699-07	D1	10	0	0	8	8	8
source-v23-6699-08	D2	10	0	0	13	13	13
source-v23-6699-09	D2	10	2	0	11	11	11
source-v23-6699-10	D1	10	2	0	7	7	7
source-v23-6699-11	D2	10	4	0	15	15	15
source-v23-6699-12	D2	10	4	0	15	15	15
source-v23-21004-01	D1	10	0	0	8	8	8
source-v23-21004-02	D2	10	0	0	10	10	10
source-v23-21004-03	D2	10	2	0	12	12	12
source-v23-21004-04	D2	10	2	0	15	15	15
source-v23-21004-05	D2	10	4	0	16	16	16
source-v23-21004-06	D2	10	4	0	14	14	14
source-v23-21004-07	D1	10	0	0	8	8	8
source-v23-21004-08	D2	10	0	0	10	10	10
source-v23-21004-09	D2	10	2	0	13	13	13
source-v23-21004-10	D2	10	2	0	13	13	13
source-v23-21004-11	D3	10	4	0	17	17	17
source-v23-21004-12	D2	10	4	0	12	12	12
source-v23-21005-01	D1	10	0	0	6	6	6
source-v23-21005-02	D1	10	0	0	8	8	8
source-v23-21005-03	D2	10	2	0	12	12	12
source-v23-21005-04	D2	10	2	0	10	10	10
source-v23-21005-05	D2	10	4	0	15	15	15
source-v23-21005-06	D3	10	4	0	18	18	18
source-v23-21005-07	D1	10	0	0	5	5	5
source-v23-21005-08	D1	10	0	0	6	6	6
source-v23-21005-09	D2	10	2	0	10	10	10
source-v23-21005-10	D2	10	2	0	13	13	13
source-v23-21005-11	D2	10	4	0	15	15	15
source-v23-21005-12	D2	10	4	0	13	13	13
source-v23-21006-01	D2	10	0	0	9	9	9
source-v23-21006-02	D1	10	0	0	8	8	8
source-v23-21006-03	D1	10	2	0	8	8	8
source-v23-21006-04	D2	10	2	0	11	11	11
source-v23-21006-05	D3	10	4	0	17	17	17

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-21006-06	D2	10	4	0	14	14	14
source-v23-21006-07	D1	10	0	0	8	8	8
source-v23-21006-08	D1	10	0	0	8	8	8
source-v23-21006-09	D2	10	2	0	11	11	11
source-v23-21006-10	D2	10	2	0	12	12	12
source-v23-21006-11	D3	10	4	0	18	18	18
source-v23-21006-12	D2	10	4	0	13	13	13
source-v23-21007-01	D2	10	0	0	9	9	9
source-v23-21007-02	D1	10	0	0	5	5	5
source-v23-21007-03	D2	10	2	0	15	15	15
source-v23-21007-04	D2	10	2	0	11	11	11
source-v23-21007-05	D3	10	4	0	19	19	19
source-v23-21007-06	D3	10	4	0	21	21	21
source-v23-21007-07	D1	10	0	0	7	7	7
source-v23-21007-08	D1	10	0	0	8	8	8
source-v23-21007-09	D2	10	2	0	11	11	11
source-v23-21007-10	D2	10	2	0	13	13	13
source-v23-21007-11	D2	10	4	0	14	14	14
source-v23-21007-12	D2	10	4	0	15	15	15
source-v23-21009-01	D1	10	0	0	7	7	7
source-v23-21009-02	D1	10	0	0	5	5	5
source-v23-21009-03	D2	10	2	0	12	12	12
source-v23-21009-04	D2	10	2	0	16	16	16
source-v23-21009-05	D3	10	4	0	17	17	17
source-v23-21009-06	D2	10	4	0	16	16	16
source-v23-21009-07	D1	10	0	0	7	7	7
source-v23-21009-08	D1	10	0	0	6	6	6
source-v23-21009-09	D2	10	2	0	16	16	16
source-v23-21009-10	D2	10	2	0	12	12	12
source-v23-21009-11	D3	10	4	0	18	18	18
source-v23-21009-12	D3	10	4	0	19	19	19
source-v23-21010-01	D2	10	0	0	10	10	10
source-v23-21010-02	D2	10	0	0	9	9	9
source-v23-21010-03	D2	10	2	0	12	12	12
source-v23-21010-04	D2	10	2	0	10	10	10
source-v23-21010-05	D2	10	4	0	12	12	12
source-v23-21010-06	D2	10	4	0	12	12	12
source-v23-21010-07	D2	10	0	0	9	9	9
source-v23-21010-08	D1	10	0	0	7	7	7
source-v23-21010-09	D1	10	2	0	6	6	6
source-v23-21010-10	D2	10	2	0	10	10	10
source-v23-21010-11	D2	10	4	0	12	12	12
source-v23-21010-12	D2	10	4	0	11	11	11
source-v23-21011-01	D1	10	0	0	7	7	7
source-v23-21011-02	D1	10	0	0	6	6	6
source-v23-21011-03	D2	10	2	0	14	14	14
source-v23-21011-04	D2	10	2	0	10	10	10
source-v23-21011-05	D2	10	4	0	14	14	14
source-v23-21011-06	D3	10	4	0	17	17	17
source-v23-21011-07	D2	10	0	0	9	9	9
source-v23-21011-08	D1	10	0	0	8	8	8
source-v23-21011-09	D2	10	2	0	12	12	12
source-v23-21011-10	D2	10	2	0	10	10	10
source-v23-21011-11	D2	10	4	0	15	15	15
source-v23-21011-12	D2	10	4	0	12	12	12
source-v23-21012-01	D2	10	0	0	16	16	16

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-21012-02	D2	10	0	0	16	16	16
source-v23-21012-03	D3	10	2	0	19	19	19
source-v23-21012-04	D3	10	2	0	20	20	20
source-v23-21012-05	D3	10	4	0	23	23	23
source-v23-21012-06	D3	10	4	0	23	23	23
source-v23-21012-07	D2	10	0	0	13	13	13
source-v23-21012-08	D2	10	0	0	15	15	17
source-v23-21012-09	D3	10	2	0	23	21	193
source-v23-21012-10	D3	10	2	0	18	16	46
source-v23-21012-11	D3	10	4	0	18	16	33
source-v23-21012-12	D3	10	4	0	20	20	20
source-v23-21014-01	D1	10	0	0	8	8	8
source-v23-21014-02	D1	10	0	0	8	8	8
source-v23-21014-03	D2	10	2	0	15	15	15
source-v23-21014-04	D2	10	2	0	15	15	15
source-v23-21014-05	D3	10	4	0	18	18	18
source-v23-21014-06	D3	10	4	0	19	19	19
source-v23-21014-07	D1	10	0	0	8	8	8
source-v23-21014-08	D2	10	0	0	12	12	12
source-v23-21014-09	D2	10	2	0	15	15	15
source-v23-21014-10	D2	10	2	0	9	9	9
source-v23-21014-11	D2	10	4	0	11	11	13
source-v23-21014-12	D2	10	4	0	13	13	13
source-v23-21016-01	D2	10	0	0	15	15	15
source-v23-21016-02	D2	10	0	0	13	13	13
source-v23-21016-03	D3	10	2	0	18	18	18
source-v23-21016-04	D3	10	2	0	19	19	19
source-v23-21016-05	D3	10	4	0	22	22	22
source-v23-21016-06	D3	10	4	0	23	23	23
source-v23-21016-07	D1	10	0	0	8	8	8
source-v23-21016-08	D2	10	0	0	15	13	34
source-v23-21016-09	D3	10	2	0	21	21	31
source-v23-21016-10	D3	10	2	0	17	17	18
source-v23-21016-11	D3	10	4	0	21	19	181
source-v23-21016-12	D3	10	4	0	23	19	101
source-v23-21017-01	D1	10	0	0	6	6	6
source-v23-21017-02	D1	10	0	0	4	4	4
source-v23-21017-03	D2	10	2	0	14	14	14
source-v23-21017-04	D2	10	2	0	14	14	14
source-v23-21017-05	D2	10	4	0	16	16	16
source-v23-21017-06	D2	10	4	0	15	15	15
source-v23-21017-07	D1	10	0	0	7	7	7
source-v23-21017-08	D2	10	0	0	10	10	10
source-v23-21017-09	D3	10	2	0	17	17	17
source-v23-21017-10	D2	10	2	0	16	16	16
source-v23-21017-11	D2	10	4	0	14	14	14
source-v23-21017-12	D2	10	4	0	14	14	14
source-v23-21018-01	D2	10	0	0	14	14	14
source-v23-21018-02	D3	10	0	0	18	18	18
source-v23-21018-03	D3	10	2	0	22	22	22
source-v23-21018-04	D2	10	2	0	14	14	14
source-v23-21018-05	D3	10	4	0	21	21	21
source-v23-21018-06	D3	10	4	0	18	18	18
source-v23-21018-07	D2	10	0	0	10	10	10
source-v23-21018-08	D1	10	0	0	7	7	9
source-v23-21018-09	D3	10	2	0	18	18	27

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-21018-10	D2	10	2	0	13	11	54
source-v23-21018-11	D3	10	4	0	19	19	25
source-v23-21018-12	D2	10	4	0	15	15	15
source-v23-21020-01	D2	10	0	0	9	9	9
source-v23-21020-02	D1	10	0	0	7	7	7
source-v23-21020-03	D2	10	2	0	10	10	10
source-v23-21020-04	D2	10	2	0	14	14	14
source-v23-21020-05	D2	10	4	0	16	16	16
source-v23-21020-06	D3	10	4	0	19	19	19
source-v23-21020-07	D2	10	0	0	10	10	10
source-v23-21020-08	D1	10	0	0	6	6	6
source-v23-21020-09	D2	10	2	0	10	10	10
source-v23-21020-10	D2	10	2	0	11	11	11
source-v23-21020-11	D2	10	4	0	9	9	9
source-v23-21020-12	D2	10	4	0	14	14	14
source-v23-21022-01	D1	10	0	0	6	6	6
source-v23-21022-02	D2	10	0	0	10	10	10
source-v23-21022-03	D2	10	2	0	15	15	15
source-v23-21022-04	D2	10	2	0	13	13	13
source-v23-21022-05	D2	10	4	0	15	15	15
source-v23-21022-06	D3	10	4	0	18	18	18
source-v23-21022-07	D1	10	0	0	8	8	8
source-v23-21022-08	D1	10	0	0	6	6	6
source-v23-21022-09	D2	10	2	0	11	11	11
source-v23-21022-10	D2	10	2	0	12	12	12
source-v23-21022-11	D2	10	4	0	15	15	15
source-v23-21022-12	D2	10	4	0	15	15	15
source-v23-21024-01	D2	10	0	0	10	10	10
source-v23-21024-02	D2	10	0	0	13	13	13
source-v23-21024-03	D3	10	2	0	18	18	18
source-v23-21024-04	D2	10	2	0	14	14	14
source-v23-21024-05	D2	10	4	0	15	15	15
source-v23-21024-06	D2	10	4	0	14	14	14
source-v23-21024-07	D2	10	0	0	13	13	13
source-v23-21024-08	D2	10	0	0	13	13	13
source-v23-21024-09	D2	10	2	0	13	13	13
source-v23-21024-10	D2	10	2	0	14	14	14
source-v23-21024-11	D2	10	4	0	15	15	15
source-v23-21024-12	D3	10	4	0	20	20	24
source-v23-21029-01	D1	10	0	0	4	4	4
source-v23-21029-02	D2	10	0	0	10	10	10
source-v23-21029-03	D2	10	2	0	11	11	11
source-v23-21029-04	D2	10	2	0	14	14	14
source-v23-21029-05	D2	10	4	0	16	16	16
source-v23-21029-06	D2	10	4	0	15	15	15
source-v23-21029-07	D2	10	0	0	10	10	10
source-v23-21029-08	D1	10	0	0	6	6	6
source-v23-21029-09	D3	10	2	0	18	18	18
source-v23-21029-10	D2	10	2	0	12	12	12
source-v23-21029-11	D2	10	4	0	16	16	16
source-v23-21029-12	D2	10	4	0	14	14	14
source-v23-21030-01	D2	10	0	0	10	10	10
source-v23-21030-02	D1	10	0	0	6	6	6
source-v23-21030-03	D2	10	2	0	16	16	16
source-v23-21030-04	D2	10	2	0	12	12	12
source-v23-21030-05	D3	10	4	0	17	17	17

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-21030-06	D3	10	4	0	19	19	19
source-v23-21030-07	D2	10	0	0	9	9	9
source-v23-21030-08	D1	10	0	0	7	7	7
source-v23-21030-09	D2	10	2	0	12	12	12
source-v23-21030-10	D2	10	2	0	16	16	16
source-v23-21030-11	D2	10	4	0	16	16	16
source-v23-21030-12	D3	10	4	0	18	18	24
source-v23-21037-01	D1	10	0	0	5	5	5
source-v23-21037-02	D2	10	0	0	13	13	13
source-v23-21037-03	D2	10	2	0	14	14	14
source-v23-21037-04	D2	10	2	0	12	12	12
source-v23-21037-05	D3	10	4	0	18	18	18
source-v23-21037-06	D3	10	4	0	19	19	19
source-v23-21037-07	D2	10	0	0	10	10	10
source-v23-21037-08	D2	10	0	0	12	12	12
source-v23-21037-09	D2	10	2	0	16	16	16
source-v23-21037-10	D2	10	2	0	12	12	12
source-v23-21037-11	D2	10	4	0	15	15	15
source-v23-21037-12	D2	10	4	0	13	13	13
source-v23-21045-01	D1	10	0	0	8	8	8
source-v23-21045-02	D2	10	0	0	10	10	10
source-v23-21045-03	D2	10	2	0	16	16	16
source-v23-21045-04	D2	10	2	0	12	12	12
source-v23-21045-05	D2	10	4	0	14	14	14
source-v23-21045-06	D2	10	4	0	14	14	14
source-v23-21045-07	D2	10	0	0	9	9	9
source-v23-21045-08	D1	10	0	0	7	7	7
source-v23-21045-09	D2	10	2	0	9	9	9
source-v23-21045-10	D2	10	2	0	16	16	16
source-v23-21045-11	D2	10	4	0	12	12	12
source-v23-21045-12	D2	10	4	0	16	16	16
source-v23-20015-01	D1	8	0	0	5	5	5
source-v23-20015-02	D1	8	0	0	6	6	6
source-v23-20015-03	D1	8	2	0	8	8	8
source-v23-20015-04	D2	8	2	0	12	12	12
source-v23-20015-05	D2	8	4	0	14	14	14
source-v23-20015-06	D2	8	4	0	13	13	13
source-v23-20015-07	D1	8	0	0	6	6	6
source-v23-20015-08	D1	8	0	0	7	7	7
source-v23-20015-09	D2	8	2	0	10	10	10
source-v23-20015-10	D2	8	2	0	12	12	12
source-v23-20015-11	D2	8	4	0	13	13	13
source-v23-20015-12	D2	8	4	0	12	12	12
source-v23-30025-01	D2	10	0	0	14	14	14
source-v23-30025-02	D2	10	0	0	15	15	15
source-v23-30025-03	D3	10	2	0	17	17	17
source-v23-30025-04	D2	10	2	0	13	13	13
source-v23-30025-05	D3	10	4	0	20	20	20
source-v23-30025-06	D3	10	4	0	18	18	18
source-v23-30025-07	D2	10	0	0	11	11	11
source-v23-30025-08	D2	10	0	0	11	11	11
source-v23-30025-09	D2	10	2	0	15	15	15
source-v23-30025-10	D2	10	2	0	11	11	11
source-v23-30025-11	D3	10	4	0	21	21	21
source-v23-30025-12	D3	10	4	0	19	19	20
source-v23-30026-01	D2	9	0	0	12	12	12

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-30026-02	D2	9	0	0	13	13	13
source-v23-30026-03	D3	9	2	0	21	21	21
source-v23-30026-04	D4	9	2	0	25	19	378
source-v23-30026-05	D2	9	4	0	15	15	15
source-v23-30026-06	D3	9	4	0	21	21	21
source-v23-30026-07	D3	9	0	0	22	12	590
source-v23-30026-08	D3	9	0	0	18	12	250
source-v23-30026-09	D3	9	2	0	24	16	2,146
source-v23-30026-10	D3	9	2	0	21	13	1,627
source-v23-30026-11	D3	9	4	0	23	13	6,383
source-v23-30026-12	D4	9	4	0	26	16	14,307
source-v23-30027-01	D2	10	0	0	13	13	13
source-v23-30027-02	D1	10	0	0	6	6	6
source-v23-30027-03	D3	10	2	0	20	20	20
source-v23-30027-04	D2	10	2	0	11	11	11
source-v23-30027-05	D3	10	4	0	18	18	18
source-v23-30027-06	D3	10	4	0	19	19	19
source-v23-30027-07	D3	10	0	0	17	17	48
source-v23-30027-08	D1	10	0	0	7	7	7
source-v23-30027-09	D1	10	2	0	6	6	6
source-v23-30027-10	D3	10	2	0	21	15	312
source-v23-30027-11	D3	10	4	0	17	17	17
source-v23-30027-12	D3	10	4	0	23	19	293
source-v23-30029-01	D3	10	0	0	19	19	19
source-v23-30029-02	D2	10	0	0	15	15	15
source-v23-30029-03	D3	10	2	0	20	20	20
source-v23-30029-04	D3	10	2	0	21	21	21
source-v23-30029-05	D3	10	4	0	24	24	24
source-v23-30029-06	D4	10	4	0	28	28	28
source-v23-30029-07	D1	10	0	0	8	8	8
source-v23-30029-08	D1	10	0	0	4	4	4
source-v23-30029-09	D2	10	2	0	16	14	199
source-v23-30029-10	D3	10	2	0	17	17	17
source-v23-30029-11	D3	10	4	0	21	21	21
source-v23-30029-12	D3	10	4	0	20	20	38
source-v23-30542-01	D1	10	0	0	3	3	3
source-v23-30542-02	D2	10	0	0	13	13	13
source-v23-30542-03	D2	10	2	0	16	16	16
source-v23-30542-04	D2	10	2	0	9	9	9
source-v23-30542-05	D3	10	4	0	21	21	21
source-v23-30542-06	D3	10	4	0	21	21	21
source-v23-30542-07	D1	10	0	0	6	6	6
source-v23-30542-08	D2	10	0	0	13	13	14
source-v23-30542-09	D2	10	2	0	15	15	15
source-v23-30542-10	D3	10	2	0	18	18	19
source-v23-30542-11	D2	10	4	0	16	16	17
source-v23-30542-12	D3	10	4	0	20	20	22
source-v23-30571-01	D3	10	0	0	17	17	17
source-v23-30571-02	D2	10	0	0	12	12	12
source-v23-30571-03	D3	10	2	0	20	20	20
source-v23-30571-04	D3	10	2	0	19	19	19
source-v23-30571-05	D4	10	4	0	27	27	36
source-v23-30571-06	D3	10	4	0	22	22	22
source-v23-30571-07	D3	10	0	0	20	20	20
source-v23-30571-08	D1	10	0	0	4	4	4
source-v23-30571-09	D2	10	2	0	15	15	15

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-30571-10	D3	10	2	0	24	24	24
source-v23-30571-11	D3	10	4	0	23	23	24
source-v23-30571-12	D3	10	4	0	23	23	23
source-v23-1682-01	D1	8	0	0	7	7	7
source-v23-1682-02	D1	8	0	0	4	4	4
source-v23-1682-03	D2	8	2	0	13	13	13
source-v23-1682-04	D2	8	2	0	14	14	14
source-v23-1682-05	D3	8	4	0	20	20	20
source-v23-1682-06	D2	8	4	0	16	16	16
source-v23-1682-07	D2	8	0	0	9	7	9
source-v23-1682-08	D1	8	0	0	5	5	5
source-v23-1682-09	D2	8	2	0	13	13	13
source-v23-1682-10	D2	8	2	0	12	12	20
source-v23-1682-11	D2	8	4	0	14	14	18
source-v23-1682-12	D2	8	4	0	16	16	25
source-v23-1775-01	D1	9	0	0	6	6	6
source-v23-1775-02	D1	9	0	0	7	7	7
source-v23-1775-03	D2	9	2	0	11	11	11
source-v23-1775-04	D1	9	2	0	8	8	8
source-v23-1775-05	D2	9	4	0	15	15	15
source-v23-1775-06	D2	9	4	0	14	14	14
source-v23-1775-07	D1	9	0	0	4	4	4
source-v23-1775-08	D1	9	0	0	7	7	7
source-v23-1775-09	D2	9	2	0	9	9	9
source-v23-1775-10	D1	9	2	0	6	6	6
source-v23-1775-11	D2	9	4	0	14	14	14
source-v23-1775-12	D2	9	4	0	11	11	11
source-v23-1808-01	D2	10	0	0	13	13	13
source-v23-1808-02	D2	10	0	0	10	10	10
source-v23-1808-03	D3	10	2	0	22	22	22
source-v23-1808-04	D3	10	2	0	18	18	18
source-v23-1808-05	D3	10	4	0	24	24	24
source-v23-1808-06	D4	10	4	0	28	28	28
source-v23-1808-07	D1	10	0	0	7	7	7
source-v23-1808-08	D1	10	0	0	7	7	7
source-v23-1808-09	D3	10	2	0	19	19	19
source-v23-1808-10	D3	10	2	0	17	17	17
source-v23-1808-11	D3	10	4	0	21	21	23
source-v23-1808-12	D3	10	4	0	19	19	19
source-v23-21321-01	D1	10	0	0	8	8	8
source-v23-21321-02	D1	10	0	0	8	8	8
source-v23-21321-03	D3	10	2	0	17	17	17
source-v23-21321-04	D2	10	2	0	16	16	16
source-v23-21321-05	D3	10	4	0	20	20	20
source-v23-21321-06	D3	10	4	0	21	21	21
source-v23-21321-07	D2	10	0	0	9	9	9
source-v23-21321-08	D1	10	0	0	7	7	7
source-v23-21321-09	D2	10	2	0	14	14	14
source-v23-21321-10	D2	10	2	0	14	14	15
source-v23-21321-11	D2	10	4	0	14	14	14
source-v23-21321-12	D2	10	4	0	16	16	16
source-v23-30181-01	D1	5	0	0	4	4	4
source-v23-30181-02	D1	5	0	0	8	8	8
source-v23-30181-03	D2	5	2	0	12	12	12
source-v23-30181-04	D2	5	2	0	10	10	10
source-v23-30181-05	D2	5	4	0	13	13	13

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-30181-06	D2	5	4	0	14	14	14
source-v23-30181-07	D1	5	0	0	3	3	3
source-v23-30181-08	D1	5	0	0	4	4	4
source-v23-30181-09	D1	5	2	0	7	7	8
source-v23-30181-10	D2	5	2	0	10	10	20
source-v23-30181-11	D2	5	4	0	12	10	14
source-v23-30181-12	D2	5	4	0	12	10	20
source-v23-30311-01	D1	10	0	0	5	5	5
source-v23-30311-02	D2	10	0	0	11	11	11
source-v23-30311-03	D4	10	2	0	25	25	25
source-v23-30311-04	D3	10	2	0	23	23	23
source-v23-30311-05	D3	10	4	0	23	23	23
source-v23-30311-06	D4	10	4	0	26	26	29
source-v23-30311-07	D2	10	0	0	13	13	13
source-v23-30311-08	D1	10	0	0	8	8	8
source-v23-30311-09	D3	10	2	0	18	18	30
source-v23-30311-10	D3	10	2	0	19	19	35
source-v23-30311-11	D3	10	4	0	21	21	60
source-v23-30311-12	D3	10	4	0	21	19	138
source-v23-31039-01	D2	10	0	0	10	10	10
source-v23-31039-02	D1	10	0	0	8	8	8
source-v23-31039-03	D2	10	2	0	11	11	11
source-v23-31039-04	D2	10	2	0	11	11	11
source-v23-31039-05	D3	10	4	0	17	17	17
source-v23-31039-06	D2	10	4	0	16	16	16
source-v23-31039-07	D2	10	0	0	13	13	13
source-v23-31039-08	D2	10	0	0	9	9	9
source-v23-31039-09	D2	10	2	0	13	13	13
source-v23-31039-10	D2	10	2	0	15	15	15
source-v23-31039-11	D2	10	4	0	16	16	16
source-v23-31039-12	D2	10	4	0	12	12	12
source-v23-31049-01	D2	9	0	0	9	9	9
source-v23-31049-02	D2	9	0	0	9	9	9
source-v23-31049-03	D2	9	2	0	15	15	15
source-v23-31049-04	D2	9	2	0	16	16	16
source-v23-31049-05	D2	9	4	0	15	15	15
source-v23-31049-06	D3	9	4	0	20	20	20
source-v23-31049-07	D1	9	0	0	7	7	7
source-v23-31049-08	D1	9	0	0	6	6	6
source-v23-31049-09	D1	9	2	0	5	5	5
source-v23-31049-10	D2	9	2	0	9	9	9
source-v23-31049-11	D2	9	4	0	13	13	13
source-v23-31049-12	D2	9	4	0	13	13	13
source-v23-31066-01	D1	10	0	0	6	6	6
source-v23-31066-02	D1	10	0	0	7	7	7
source-v23-31066-03	D2	10	2	0	16	16	16
source-v23-31066-04	D2	10	2	0	11	11	11
source-v23-31066-05	D3	10	4	0	21	21	21
source-v23-31066-06	D3	10	4	0	24	24	24
source-v23-31066-07	D1	10	0	0	8	8	8
source-v23-31066-08	D1	10	0	0	5	5	5
source-v23-31066-09	D2	10	2	0	15	15	15
source-v23-31066-10	D2	10	2	0	15	15	15
source-v23-31066-11	D3	10	4	0	18	18	18
source-v23-31066-12	D3	10	4	0	20	20	20
source-v23-31092-01	D2	10	0	0	11	11	11

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-31092-02	D1	10	0	0	8	8	8
source-v23-31092-03	D3	10	2	0	18	18	18
source-v23-31092-04	D1	10	2	0	8	8	8
source-v23-31092-05	D3	10	4	0	17	17	17
source-v23-31092-06	D2	10	4	0	16	16	16
source-v23-31092-07	D1	10	0	0	5	5	5
source-v23-31092-08	D2	10	0	0	13	13	13
source-v23-31092-09	D2	10	2	0	12	12	12
source-v23-31092-10	D2	10	2	0	11	11	11
source-v23-31092-11	D3	10	4	0	17	17	17
source-v23-31092-12	D2	10	4	0	15	15	15
source-v23-3181-01	D1	6	0	0	8	8	8
source-v23-3181-02	D1	6	0	0	8	8	8
source-v23-3181-03	D2	6	2	0	13	13	13
source-v23-3181-04	D2	6	2	0	12	12	12
source-v23-3181-05	D2	6	4	0	16	16	16
source-v23-3181-06	D2	6	4	0	16	16	16
source-v23-3181-07	D1	6	0	0	5	5	5
source-v23-3181-08	D2	6	0	0	11	9	14
source-v23-3181-09	D1	6	2	0	6	6	6
source-v23-3181-10	D1	6	2	0	8	8	8
source-v23-3181-11	D2	6	4	0	10	10	10
source-v23-3181-12	D2	6	4	0	13	13	13
source-v23-6745-01	D2	10	0	0	13	13	13
source-v23-6745-02	D1	10	0	0	6	6	6
source-v23-6745-03	D3	10	2	0	19	19	19
source-v23-6745-04	D2	10	2	0	16	16	16
source-v23-6745-05	D3	10	4	0	18	18	18
source-v23-6745-06	D3	10	4	0	20	20	20
source-v23-6745-07	D2	10	0	0	11	11	11
source-v23-6745-08	D2	10	0	0	12	12	12
source-v23-6745-09	D2	10	2	0	13	13	13
source-v23-6745-10	D2	10	2	0	13	13	13
source-v23-6745-11	D2	10	4	0	16	16	16
source-v23-6745-12	D3	10	4	0	20	20	20
source-v23-21303-01	D1	9	0	0	8	8	8
source-v23-21303-02	D1	9	0	0	6	6	6
source-v23-21303-03	D2	9	2	0	14	14	14
source-v23-21303-04	D2	9	2	0	12	12	12
source-v23-21303-05	D2	9	4	0	16	16	16
source-v23-21303-06	D3	9	4	0	17	17	17
source-v23-21303-07	D1	9	0	0	8	8	8
source-v23-21303-08	D1	9	0	0	7	7	7
source-v23-21303-09	D2	9	2	0	11	11	11
source-v23-21303-10	D2	9	2	0	11	11	11
source-v23-21303-11	D2	9	4	0	15	15	15
source-v23-21303-12	D2	9	4	0	14	14	14
source-v23-1489-01	D1	5	0	0	4	4	4
source-v23-1489-02	D1	5	0	0	5	5	5
source-v23-1489-03	D1	5	2	0	7	7	7
source-v23-1489-04	D1	5	2	0	8	8	8
source-v23-1489-05	D2	5	4	0	14	14	14
source-v23-1489-06	D2	5	4	0	15	15	15
source-v23-1489-07	D1	5	0	0	4	4	4
source-v23-1489-08	D1	5	0	0	2	2	2
source-v23-1489-09	D1	5	2	0	8	8	8

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-1489-10	D1	5	2	0	6	6	6
source-v23-1489-11	D2	5	4	0	12	12	12
source-v23-1489-12	D2	5	4	0	12	12	12
source-v23-1572-01	D1	10	0	0	8	8	8
source-v23-1572-02	D2	10	0	0	10	10	10
source-v23-1572-03	D2	10	2	0	10	10	10
source-v23-1572-04	D2	10	2	0	9	9	9
source-v23-1572-05	D2	10	4	0	11	11	11
source-v23-1572-06	D2	10	4	0	14	14	14
source-v23-1572-07	D1	10	0	0	5	5	5
source-v23-1572-08	D1	10	0	0	8	8	8
source-v23-1572-09	D1	10	2	0	5	5	5
source-v23-1572-10	D1	10	2	0	7	7	7
source-v23-1572-11	D2	10	4	0	12	12	12
source-v23-1572-12	D2	10	4	0	11	11	11
source-v23-1633-01	D1	8	0	0	8	8	8
source-v23-1633-02	D1	8	0	0	5	5	5
source-v23-1633-03	D1	8	2	0	8	8	8
source-v23-1633-04	D1	8	2	0	5	5	5
source-v23-1633-05	D2	8	4	0	14	14	14
source-v23-1633-06	D2	8	4	0	15	15	15
source-v23-1633-07	D1	8	0	0	3	3	3
source-v23-1633-08	D1	8	0	0	3	3	3
source-v23-1633-09	D1	8	2	0	7	7	7
source-v23-1633-10	D1	8	2	0	8	8	8
source-v23-1633-11	D2	8	4	0	10	10	10
source-v23-1633-12	D1	8	4	0	7	7	7
source-v23-31084-01	D1	8	0	0	3	3	3
source-v23-31084-02	D2	8	0	0	11	11	11
source-v23-31084-03	D2	8	2	0	12	12	12
source-v23-31084-04	D2	8	2	0	10	10	10
source-v23-31084-05	D3	8	4	0	17	17	17
source-v23-31084-06	D2	8	4	0	15	15	15
source-v23-31084-07	D1	8	0	0	4	4	4
source-v23-31084-08	D1	8	0	0	7	7	7
source-v23-31084-09	D2	8	2	0	10	10	10
source-v23-31084-10	D2	8	2	0	10	10	11
source-v23-31084-11	D2	8	4	0	12	12	12
source-v23-31084-12	D2	8	4	0	11	11	11
source-v23-3179-01	D3	10	0	0	17	17	17
source-v23-3179-02	D3	10	0	0	17	17	17
source-v23-3179-03	D3	10	2	0	21	21	21
source-v23-3179-04	D3	10	2	0	17	17	17
source-v23-3179-05	D3	10	4	0	18	18	18
source-v23-3179-06	D3	10	4	0	19	19	19
source-v23-3179-07	D2	10	0	0	14	14	14
source-v23-3179-08	D2	10	0	0	11	11	11
source-v23-3179-09	D2	10	2	0	13	13	19
source-v23-3179-10	D2	10	2	0	14	14	14
source-v23-3179-11	D3	10	4	0	20	20	20
source-v23-3179-12	D3	10	4	0	21	21	21
source-v23-6678-01	D2	9	0	0	10	10	10
source-v23-6678-02	D2	9	0	0	10	10	10
source-v23-6678-03	D2	9	2	0	13	13	13
source-v23-6678-04	D2	9	2	0	13	13	13
source-v23-6678-05	D3	9	4	0	18	18	18

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-6678-06	D3	9	4	0	19	19	19
source-v23-6678-07	D2	9	0	0	9	9	9
source-v23-6678-08	D1	9	0	0	8	8	8
source-v23-6678-09	D2	9	2	0	9	9	10
source-v23-6678-10	D2	9	2	0	11	11	11
source-v23-6678-11	D2	9	4	0	15	15	15
source-v23-6678-12	D2	9	4	0	13	13	14
source-v23-6682-01	D1	10	0	0	8	8	8
source-v23-6682-02	D1	10	0	0	6	6	6
source-v23-6682-03	D2	10	2	0	15	15	15
source-v23-6682-04	D2	10	2	0	14	14	14
source-v23-6682-05	D4	10	4	0	27	27	27
source-v23-6682-06	D4	10	4	0	26	26	27
source-v23-6682-07	D2	10	0	0	10	10	10
source-v23-6682-08	D1	10	0	0	7	7	7
source-v23-6682-09	D2	10	2	0	12	12	12
source-v23-6682-10	D2	10	2	0	16	16	16
source-v23-6682-11	D3	10	4	0	19	19	67
source-v23-6682-12	D3	10	4	0	18	18	104
source-v23-6690-01	D3	10	0	0	17	17	17
source-v23-6690-02	D3	10	0	0	18	18	18
source-v23-6690-03	D3	10	2	0	18	18	18
source-v23-6690-04	D3	10	2	0	21	21	21
source-v23-6690-05	D3	10	4	0	18	18	18
source-v23-6690-06	D3	10	4	0	20	20	20
source-v23-6690-07	D2	10	0	0	13	13	23
source-v23-6690-08	D2	10	0	0	13	13	20
source-v23-6690-09	D3	10	2	0	17	17	25
source-v23-6690-10	D2	10	2	0	16	16	32
source-v23-6690-11	D2	10	4	0	13	13	13
source-v23-6690-12	D3	10	4	0	18	18	82
source-v23-10248-01	D1	8	0	0	6	6	6
source-v23-10248-02	D1	8	0	0	7	7	7
source-v23-10248-03	D2	8	2	0	10	10	10
source-v23-10248-04	D2	8	2	0	11	11	11
source-v23-10248-05	D2	8	4	0	15	15	15
source-v23-10248-06	D2	8	4	0	14	14	14
source-v23-10248-07	D1	8	0	0	1	1	1
source-v23-10248-08	D1	8	0	0	4	4	4
source-v23-10248-09	D1	8	2	0	6	6	6
source-v23-10248-10	D1	8	2	0	7	7	7
source-v23-10248-11	D2	8	4	0	11	11	11
source-v23-10248-12	D2	8	4	0	9	9	9
source-v23-10258-01	D1	5	0	0	1	1	1
source-v23-10258-02	D1	5	0	0	5	5	5
source-v23-10258-03	D1	5	2	0	8	8	8
source-v23-10258-04	D2	5	2	0	10	10	10
source-v23-10258-05	D2	5	4	0	10	10	10
source-v23-10258-06	D2	5	4	0	11	11	11
source-v23-10258-07	D1	5	0	0	3	3	3
source-v23-10258-08	D1	5	0	0	4	4	4
source-v23-10258-09	D1	5	2	0	7	7	7
source-v23-10258-10	D1	5	2	0	8	8	8
source-v23-10258-11	D2	5	4	0	9	9	9
source-v23-10258-12	D2	5	4	0	11	11	11
source-v23-10271-01	D1	5	0	0	3	3	3

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-10271-02	D1	5	0	0	3	3	3
source-v23-10271-03	D2	5	2	0	13	13	13
source-v23-10271-04	D2	5	2	0	9	9	9
source-v23-10271-05	D2	5	4	0	14	14	14
source-v23-10271-06	D2	5	4	0	15	15	15
source-v23-10271-07	D1	5	0	0	4	4	4
source-v23-10271-08	D1	5	0	0	4	4	4
source-v23-10271-09	D2	5	2	0	10	10	10
source-v23-10271-10	D2	5	2	0	9	9	9
source-v23-10271-11	D2	5	4	0	11	11	11
source-v23-10271-12	D2	5	4	0	14	14	14
source-v23-31037-01	D2	10	0	0	9	9	9
source-v23-31037-02	D1	10	0	0	7	7	7
source-v23-31037-03	D2	10	2	0	14	14	14
source-v23-31037-04	D2	10	2	0	12	12	12
source-v23-31037-05	D3	10	4	0	18	18	18
source-v23-31037-06	D3	10	4	0	18	18	18
source-v23-31037-07	D1	10	0	0	6	6	6
source-v23-31037-08	D1	10	0	0	8	8	8
source-v23-31037-09	D2	10	2	0	12	12	12
source-v23-31037-10	D2	10	2	0	14	14	14
source-v23-31037-11	D2	10	4	0	15	15	15
source-v23-31037-12	D2	10	4	0	14	14	14
source-v23-31040-01	D1	10	0	0	6	6	6
source-v23-31040-02	D2	10	0	0	11	11	11
source-v23-31040-03	D2	10	2	0	12	12	12
source-v23-31040-04	D2	10	2	0	12	12	12
source-v23-31040-05	D2	10	4	0	13	13	13
source-v23-31040-06	D3	10	4	0	18	18	18
source-v23-31040-07	D1	10	0	0	8	8	8
source-v23-31040-08	D1	10	0	0	6	6	6
source-v23-31040-09	D2	10	2	0	10	10	10
source-v23-31040-10	D2	10	2	0	9	9	9
source-v23-31040-11	D3	10	4	0	17	17	17
source-v23-31040-12	D2	10	4	0	12	12	12
source-v23-31070-01	D2	10	0	0	9	9	9
source-v23-31070-02	D1	10	0	0	6	6	6
source-v23-31070-03	D2	10	2	0	14	14	14
source-v23-31070-04	D2	10	2	0	12	12	12
source-v23-31070-05	D3	10	4	0	23	23	23
source-v23-31070-06	D3	10	4	0	22	22	22
source-v23-31070-07	D1	10	0	0	5	5	5
source-v23-31070-08	D1	10	0	0	7	7	7
source-v23-31070-09	D2	10	2	0	14	14	14
source-v23-31070-10	D1	10	2	0	8	8	8
source-v23-31070-11	D3	10	4	0	17	17	17
source-v23-31070-12	D3	10	4	0	19	19	19
source-v23-31072-01	D1	8	0	0	5	5	5
source-v23-31072-02	D2	8	0	0	11	11	11
source-v23-31072-03	D2	8	2	0	13	13	13
source-v23-31072-04	D2	8	2	0	14	14	14
source-v23-31072-05	D3	8	4	0	18	18	18
source-v23-31072-06	D3	8	4	0	17	17	17
source-v23-31072-07	D1	8	0	0	7	7	7
source-v23-31072-08	D1	8	0	0	8	8	8
source-v23-31072-09	D2	8	2	0	11	11	11

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-31072-10	D2	8	2	0	15	15	15
source-v23-31072-11	D3	8	4	0	19	19	19
source-v23-31072-12	D3	8	4	0	17	17	17
source-v23-3221-01	D2	10	0	0	12	12	12
source-v23-3221-02	D2	10	0	0	13	13	13
source-v23-3221-03	D2	10	2	0	13	13	13
source-v23-3221-04	D2	10	2	0	15	15	15
source-v23-3221-05	D2	10	4	0	15	15	15
source-v23-3221-06	D3	10	4	0	19	19	19
source-v23-3221-07	D1	10	0	0	6	6	6
source-v23-3221-08	D2	10	0	0	10	10	10
source-v23-3221-09	D2	10	2	0	11	11	11
source-v23-3221-10	D2	10	2	0	10	10	10
source-v23-3221-11	D3	10	4	0	18	18	18
source-v23-3221-12	D2	10	4	0	15	15	15
source-v23-6753-01	D2	10	0	0	14	14	14
source-v23-6753-02	D1	10	0	0	5	5	5
source-v23-6753-03	D3	10	2	0	18	18	18
source-v23-6753-04	D3	10	2	0	22	22	22
source-v23-6753-05	D2	10	4	0	13	13	13
source-v23-6753-06	D3	10	4	0	22	22	22
source-v23-6753-07	D2	10	0	0	14	14	14
source-v23-6753-08	D2	10	0	0	11	11	11
source-v23-6753-09	D2	10	2	0	13	13	13
source-v23-6753-10	D2	10	2	0	13	13	13
source-v23-6753-11	D3	10	4	0	21	21	21
source-v23-6753-12	D3	10	4	0	17	17	17
source-v23-10158-01	D1	5	0	0	4	4	4
source-v23-10158-02	D1	5	0	0	3	3	3
source-v23-10158-03	D2	5	2	0	9	9	9
source-v23-10158-04	D1	5	2	0	6	6	6
source-v23-10158-05	D2	5	4	0	13	13	13
source-v23-10158-06	D2	5	4	0	12	12	12
source-v23-10158-07	D1	5	0	0	2	2	2
source-v23-10158-08	D1	5	0	0	2	2	2
source-v23-10158-09	D1	5	2	0	6	6	6
source-v23-10158-10	D1	5	2	0	6	6	6
source-v23-10158-11	D1	5	4	0	7	7	7
source-v23-10158-12	D1	5	4	0	5	5	5
source-v23-10194-01	D1	8	0	0	7	7	7
source-v23-10194-02	D1	8	0	0	6	6	6
source-v23-10194-03	D1	8	2	0	8	8	8
source-v23-10194-04	D2	8	2	0	10	10	10
source-v23-10194-05	D2	8	4	0	14	14	14
source-v23-10194-06	D2	8	4	0	12	12	12
source-v23-10194-07	D1	8	0	0	7	7	7
source-v23-10194-08	D1	8	0	0	7	7	7
source-v23-10194-09	D2	8	2	0	10	10	10
source-v23-10194-10	D2	8	2	0	10	10	10
source-v23-10194-11	D2	8	4	0	14	14	14
source-v23-10194-12	D3	8	4	0	17	17	17
source-v23-10233-01	D2	10	0	0	9	9	9
source-v23-10233-02	D1	10	0	0	7	7	7
source-v23-10233-03	D2	10	2	0	12	12	12
source-v23-10233-04	D2	10	2	0	14	14	14
source-v23-10233-05	D3	10	4	0	20	20	20

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-10233-06	D3	10	4	0	18	18	18
source-v23-10233-07	D2	10	0	0	9	9	9
source-v23-10233-08	D2	10	0	0	12	12	12
source-v23-10233-09	D2	10	2	0	15	15	15
source-v23-10233-10	D2	10	2	0	11	11	11
source-v23-10233-11	D3	10	4	0	19	19	19
source-v23-10233-12	D2	10	4	0	14	14	14
source-v23-10241-01	D2	10	0	0	11	11	11
source-v23-10241-02	D2	10	0	0	10	10	10
source-v23-10241-03	D2	10	2	0	16	16	16
source-v23-10241-04	D2	10	2	0	13	13	13
source-v23-10241-05	D3	10	4	0	19	19	19
source-v23-10241-06	D3	10	4	0	21	21	21
source-v23-10241-07	D2	10	0	0	13	13	13
source-v23-10241-08	D2	10	0	0	9	9	9
source-v23-10241-09	D3	10	2	0	18	18	19
source-v23-10241-10	D2	10	2	0	14	14	16
source-v23-10241-11	D3	10	4	0	17	17	17
source-v23-10241-12	D2	10	4	0	16	16	16
source-v23-31045-01	D3	10	0	0	19	19	19
source-v23-31045-02	D3	10	0	0	17	17	17
source-v23-31045-03	D3	10	2	0	19	19	19
source-v23-31045-04	D2	10	2	0	15	15	15
source-v23-31045-05	D4	10	4	0	25	25	25
source-v23-31045-06	D4	10	4	0	25	25	25
source-v23-31045-07	D2	10	0	0	14	12	16
source-v23-31045-08	D2	10	0	0	15	15	24
source-v23-31045-09	D3	10	2	0	19	19	19
source-v23-31045-10	D3	10	2	0	19	19	25
source-v23-31045-11	D3	10	4	0	20	20	20
source-v23-31045-12	D3	10	4	0	22	22	24
source-v23-4018-01	D1	5	0	0	5	5	5
source-v23-4018-02	D1	5	0	0	7	7	7
source-v23-4018-03	D2	5	2	0	11	11	11
source-v23-4018-04	D2	5	2	0	10	10	10
source-v23-4018-05	D2	5	4	0	15	15	15
source-v23-4018-06	D2	5	4	0	15	15	15
source-v23-4018-07	D1	5	0	0	8	8	8
source-v23-4018-08	D1	5	0	0	6	6	6
source-v23-4018-09	D1	5	2	0	8	8	8
source-v23-4018-10	D1	5	2	0	8	8	8
source-v23-4018-11	D2	5	4	0	10	10	10
source-v23-4018-12	D2	5	4	0	11	11	11
source-v23-6698-01	D1	10	0	0	6	6	6
source-v23-6698-02	D2	10	0	0	9	9	9
source-v23-6698-03	D2	10	2	0	12	12	12
source-v23-6698-04	D2	10	2	0	9	9	9
source-v23-6698-05	D2	10	4	0	16	16	16
source-v23-6698-06	D3	10	4	0	17	17	17
source-v23-6698-07	D1	10	0	0	4	4	4
source-v23-6698-08	D1	10	0	0	6	6	6
source-v23-6698-09	D2	10	2	0	11	11	11
source-v23-6698-10	D2	10	2	0	11	11	11
source-v23-6698-11	D2	10	4	0	12	12	12
source-v23-6698-12	D2	10	4	0	10	10	10
source-v23-1632-01	D1	5	0	0	5	5	5

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-1632-02	D1	5	0	0	6	6	6
source-v23-1632-03	D2	5	2	0	12	12	14
source-v23-1632-04	D2	5	2	0	14	14	16
source-v23-1632-05	D2	5	4	0	14	14	15
source-v23-1632-06	D3	5	4	0	17	17	18
source-v23-1632-07	D1	5	0	0	3	3	3
source-v23-1632-08	D1	5	0	0	3	3	4
source-v23-1632-09	D1	5	2	0	7	7	10
source-v23-1632-10	D2	5	2	0	9	9	13
source-v23-1632-11	D2	5	4	0	14	10	110
source-v23-1632-12	D2	5	4	0	13	11	21
house-roof-matched-1-context-1-01	D3	5	0	0	17	17	20
house-roof-matched-1-context-0-01	D3	5	0	0	17	17	21
house-roof-matched-1-context-1-02	D2	5	0	0	14	14	15
house-roof-matched-1-context-0-02	D2	5	0	0	14	14	16
source-10014-matched-0-context-0-01	D3	10	1	0	18	18	18
source-10014-matched-0-context-1-01	D3	10	1	0	18	18	18
source-10014-matched-0-context-1-02	D3	10	1	0	19	19	19
source-10014-matched-0-context-0-02	D3	10	1	0	19	19	19
source-10128-matched-0-context-0-01	D2	10	1	0	13	13	13
source-10128-matched-0-context-1-01	D2	10	1	0	13	13	13
source-10128-matched-0-context-0-02	D2	10	1	0	13	13	13
source-10128-matched-0-context-1-02	D2	10	1	0	13	13	13
source-10036-matched-0-context-1-01	D3	10	0	0	18	18	18
source-10036-matched-0-context-0-01	D3	10	0	0	18	18	18
source-10036-matched-0-context-0-02	D2	10	0	0	16	16	16
source-10036-matched-0-context-1-02	D2	10	0	0	16	16	16
source-10036-matched-2-context-1-01	D3	10	0	0	17	17	17
source-10036-matched-2-context-0-01	D3	10	0	0	17	17	17
source-10036-matched-2-context-1-02	D3	10	0	0	19	19	20
source-10036-matched-2-context-0-02	D3	10	0	0	19	19	19
source-v23-10037-matched-0-context-0-01	D1	10	1	0	7	7	7
source-v23-10037-matched-0-context-1-01	D1	10	1	0	7	7	7
source-v23-10037-matched-0-context-0-02	D2	10	1	0	11	11	11
source-v23-10037-matched-0-context-1-02	D2	10	1	0	11	11	11
source-v23-1472-matched-2-context-0-01	D2	10	0	0	16	16	16
source-v23-1472-matched-2-context-1-01	D2	10	0	0	16	16	16
source-v23-1472-matched-2-context-1-02	D2	10	0	0	15	15	15
source-v23-1472-matched-2-context-0-02	D2	10	0	0	15	15	15
source-v23-31025-matched-0-context-1-01	D2	10	0	0	15	15	15
source-v23-31025-matched-0-context-0-01	D2	10	0	0	15	15	15
source-v23-31025-matched-0-context-0-02	D2	10	0	0	12	12	12
source-v23-31025-matched-0-context-1-02	D2	10	0	0	12	12	12
source-v23-31048-matched-0-context-0-01	D3	10	0	0	24	20	50
source-v23-31048-matched-0-context-1-01	D3	10	0	0	24	20	62
source-v23-31048-matched-0-context-1-02	D3	10	0	0	24	18	76
source-v23-31048-matched-0-context-0-02	D3	10	0	0	24	18	52
source-v23-31048-matched-1-context-0-01	D3	10	0	0	24	18	67
source-v23-31048-matched-1-context-1-01	D3	10	0	0	24	18	76
source-v23-31048-matched-1-context-1-02	D3	10	0	0	22	18	33
source-v23-31048-matched-1-context-0-02	D3	10	0	0	22	18	29
source-v23-31048-matched-2-context-0-01	D3	10	0	0	24	20	37
source-v23-31048-matched-2-context-1-01	D3	10	0	0	24	20	42
source-v23-31048-matched-2-context-1-02	D3	10	0	0	24	18	52
source-v23-31048-matched-2-context-0-02	D3	10	0	0	24	18	45
source-v23-31048-matched-3-context-0-01	D4	10	0	0	26	20	79

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-v23-31048-matched-3-context-1-01	D4	10	0	0	26	20	91
source-v23-31048-matched-3-context-0-02	D3	10	0	0	20	18	22
source-v23-31048-matched-3-context-1-02	D3	10	0	0	20	18	22
source-v23-6683-matched-2-context-1-01	D3	10	0	0	17	17	17
source-v23-6683-matched-2-context-0-01	D3	10	0	0	17	17	17
source-v23-6683-matched-2-context-0-02	D2	10	0	0	13	13	13
source-v23-6683-matched-2-context-1-02	D2	10	0	0	13	13	13
source-v23-6683-matched-3-context-1-01	D2	10	0	0	15	15	15
source-v23-6683-matched-3-context-0-01	D2	10	0	0	15	15	15
source-v23-6683-matched-3-context-0-02	D2	10	0	0	12	12	12
source-v23-6683-matched-3-context-1-02	D2	10	0	0	12	12	12
source-v23-21018-matched-0-context-1-01	D3	10	1	0	17	17	17
source-v23-21018-matched-0-context-0-01	D3	10	1	0	17	17	17
source-v23-21018-matched-0-context-1-02	D3	10	1	0	17	17	17
source-v23-21018-matched-0-context-0-02	D3	10	1	0	17	17	17
source-v23-31084-matched-1-context-0-01	D2	8	1	0	12	12	12
source-v23-31084-matched-1-context-1-01	D2	8	1	0	12	12	12
source-v23-31084-matched-1-context-1-02	D2	8	1	0	11	11	11
source-v23-31084-matched-1-context-0-02	D2	8	1	0	11	11	11
source-v23-10241-matched-1-context-1-01	D2	10	0	0	14	14	14
source-v23-10241-matched-1-context-0-01	D2	10	0	0	14	14	14
source-v23-10241-matched-1-context-1-02	D2	10	0	0	16	16	16
source-v23-10241-matched-1-context-0-02	D2	10	0	0	16	16	16
source-v23-10241-matched-2-context-1-01	D3	10	0	0	22	22	22
source-v23-10241-matched-2-context-0-01	D3	10	0	0	22	22	22
source-v23-10241-matched-2-context-1-02	D3	10	0	0	22	22	22
source-v23-10241-matched-2-context-0-02	D3	10	0	0	22	22	22